

ObjektinisUzdavinys v3.0

3.0

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v3.0 relisas: Custom Vector<T> Konteineris

Vector<T> Spartos Palyginimas su std::vector

push_back() Operacijos Spartos Testas

Element skaiius	std::vector (s)	Vector<int> (s)
1000	0.04200 s	0.04292 s
10000	0.24998 s	0.31445 s
100000	2.31416 s	2.91781 s
1000000	23.71250 s	28.25032 s

Testas: std::vector vs Vector<int> push_back()

Optimizavimo Pastabos

Vector<T> Realizacijos Optimizavimas:

1. Reallokacijos Strategija: Vector naudoja geometrin augimo strategij (2x multiplier), kuri sumaina reallokacijas
 - Naujos talpos formula: $\text{new_capacity} = (_capacity == 0) ? 1 : _capacity \cdot 2$
 - is metodas sumaina $O(n^2)$ $O(n)$ amortizuot laiko sudtingum
2. Move Semantika: push_back(T&&) overload naudoja move semantik greitam itekli permimui
 - Kopijuoti brangs objektai perkeliama vietoj kopijuose
 - Efektyvu: `v.push_back(std::move(tempObj))`
3. Reserve Optimizavimas: Naudotojo optimizavimas per reserve()

```
Vector<Studentas> v;
v.reserve(100000); // I anksto alokuoti atmint - ivengiame 17 reallokacij
for (int i = 0; i < 100000; ++i) {
    v.push_back(Studentas(...)); // Dabar greitai - nereallokuojama
}
```
4. Iteratoriai: Random access iteratoriai $O(1)$ snaudos
 - begin(), end(), rbegin(), rend() veikia per $O(1)$

- Leisti range-based for ciklams ir std algoritams
5. Memory Layout: Isaugomas contiguous atmintis layout
 - data() grina raw pointer
 - Suderinamumas su C API ir SIMD operacijomis
 6. CMake Release Optimizacijos:
 - Kompiliavimas su /O2 optimizacija (MSVC Release)
 - Nra runtime debug checks /RTC1

Vector<T> API Naudojimo Pavyzdiai

1. Pagrindins Operacijos - push_back() ir pop_back()

```
#include "Vector.h"

Vector<int> v;

// push_back() - pridjimas gal
v.push_back(10);
v.push_back(20);
v.push_back(30);

std::cout << "Dydis: " << v.size() << "\n"; // Iveda: 3

// pop_back() - alinimas i galo
v.pop_back(); // alina 30

// Prieiga prie element
std::cout << "Pirmas: " << v[0] << "\n"; // Iveda: 10
std::cout << "Paskutinis: " << v.back() << "\n"; // Iveda: 20
```

2. Konstruktoriai ir Inicijalizacija

```
// Default konstruktorius
Vector<double> v1;

// Konstruktorius su dydiu
Vector<int> v2(5); // 5 int element, reikms 0

// Konstruktorius su dydiu ir reikme
Vector<std::string> v3(3, "Sveiki"); // 3 "Sveiki" eiluts

// Copy konstruktorius
Vector<int> v4 = v1; // Deep copy

// Move konstruktorius
Vector<int> v5 = std::move(v2); // Itekliai perkelti i v2 v5
```

3. Reservavimas ir Dydio Valdymas

```
Vector<int> v;
v.reserve(100); // Iankstinis atmintis rezervavimas

std::cout << "Talpa: " << v.capacity() << "\n"; // Iveda: 100
std::cout << "Dydis: " << v.size() << "\n"; // Iveda: 0

// Nustatyti konkret dyd
v.resize(50); // Dabar dydis = 50, nauji elementai = 0

// Sumainti i dalies nereikaling atmint
v.shrink_to_fit(); // capacity() == size()
```

4. Iteratoriai ir Iteravimas

```
Vector<int> v = {1, 2, 3, 4, 5};

// Range-based for ciklas
for (int elem : v) {
    std::cout << elem << " "; // Iveda: 1 2 3 4 5
}

// Tiesioginis iteratoriaus naudojimas
for (auto it = v.begin(); it != v.end(); ++it) {
```

```

    std::cout << *it << " ";
}

// Atgaline tvarka
for (auto it = v.rbegin(); it != v.rend(); ++it) {
    std::cout << *it << " "; // Iveda: 5 4 3 2 1
}

```

5. insert() ir erase() Operacijos

```

Vector<int> v = {1, 2, 3, 5};

// Djimas konkreti padt
auto it = v.begin() + 3;
v.insert(it, 4); // Dabar: {1, 2, 3, 4, 5}

// alinimas i konkreios padties
v.erase(v.begin() + 1); // alina 2, dabar: {1, 3, 4, 5}

// alinimas diapazono
v.erase(v.begin() + 1, v.begin() + 3); // alina {3, 4}, dabar: {1, 5}

```

6. assign() - Perklimas i Iteratoriaus Diapazono

```

Vector<int> v1;
std::vector<int> std_v = {10, 20, 30, 40};

// Assign i std::vector iteratori
v1.assign(std_v.begin(), std_v.end());
// Dabar v1 = {10, 20, 30, 40}

// Arba i kito Vector<int>
Vector<int> v2;
v2.assign(v1.begin(), v1.end());
// Dabar v2 = {10, 20, 30, 40}

```

7. emplace_back() - Vietos Konstravimas

```

struct Studentas {
    std::string vardas;
    int vidurkis;

    Studentas(const std::string& v, int m)
        : vardas(v), vidurkis(m) {}
};

Vector<Studentas> grup;

// emplace_back() - konstravimas tiesiog vektoriuje (be kopij)
grup.emplace_back("Jonas", 8);
grup.emplace_back("Marija", 9);

std::cout << "Pirmas studentas: " << grup[0].vardas << "\n";

```

8. Palyginimai ir Operatoriai

```

Vector<int> v1 = {1, 2, 3};
Vector<int> v2 = {1, 2, 3};
Vector<int> v3 = {1, 2, 4};

// Lygybs operatorius
if (v1 == v2) {
    std::cout << "v1 ir v2 lygs\n"; // Spausdina
}

// Nelygumo operatorius
if (v1 != v3) {
    std::cout << "v1 ir v3 nelygs\n"; // Spausdina
}

// Maiau nei operatorius
if (v1 < v3) {
    std::cout << "v1 < v3 (leksikografin slyga)\n";
}

```

Studentas Integravimas su Vector<T>

Implementacija

Klas `Studentas` (i v1.5) buvo integruota su custom `Vector<T>` kontaineriu:

Prie (v2.0 - su std::vector):

```
class Studentas : public Zmogus {
private:
    std::vector<int> paz_; // Paymiai
    int egz_;
    double rez_;
    double med_;
    // ...
};
```

Dabar (v3.0 - su Vector<T>):

```
class Studentas : public Zmogus {
private:
    Vector<int> paz_; // Custom Vector konteineris
    int egz_;
    double rez_;
    double med_;
    // ...
};
```

Metodai Integruoti su Vector

Paymi Valdymas:

```
// Paymio pridjimas
void addPaz(int paz) {
    paz_.push_back(paz);
}

// Paymi kiekis
int getPazCount() const {
    return paz_.size();
}

// Grti paymi vektori (const)
const Vector<int>& getPaz() const {
    return paz_;
}

// Paymi ataukimas
void clearPaz() {
    paz_.clear();
}

// Atmintis rezervavimas (optimizavimas)
void reservePaz(int kiekis) {
    paz_.reserve(kiekis);
}

// Paymi nustatymas i Vector
void setPaz(const Vector<int>& nauji_paz) {
    paz_ = nauji_paz;
}
```

Statistins Funkcijos su Vector

Vidutinio Skaiiavimas:

```
double vidurkis(const Vector<int>& paz) {
    if (paz.size() == 0) return 0.0;
    double suma = 0.0;
    for (int p : paz) suma += p;
    return suma / paz.size();
}
```

Medianos Skaiiavimas:

```
int mediana(Vector<int> paz) { // Kopija dl riavimo
    if (paz.size() == 0) return 0;
    std::sort(paz.begin(), paz.end());
    if (paz.size() % 2 == 1) {
        return paz[paz.size() / 2];
    }
    return (paz[paz.size() / 2 - 1] + paz[paz.size() / 2]) / 2;
}
```

Rule of Five su Vector

Copy Semantika:

```
Studentas::Studentas(const Studentas& other)
    : Zmogus(other), paz_(other.paz_), egz_(other.egz_),
      rez_(other.rez_), med_(other.med_) {}

Studentas& Studentas::operator=(const Studentas& other) {
```

```

if (this != &other) {
    Zmogus::operator=(other);
    paz_ = other.paz_; // Vector copy assignment
    egz_ = other.egz_;
    rez_ = other.rez_;
    med_ = other.med_;
}
return *this;
}

```

Move Semantika:

```

Studentas::Studentas(Studentas&& other) noexcept
: Zmogus(std::move(other)), paz_(std::move(other.paz_)),
  egz_(other.egz_), rez_(other.rez_), med_(other.med_) {}

Studentas& Studentas::operator=(Studentas&& other) noexcept {
    if (this != &other) {
        Zmogus::operator=(std::move(other));
        paz_ = std::move(other.paz_); // Vector move assignment
        egz_ = other.egz_;
        rez_ = other.rez_;
        med_ = other.med_;
    }
    return *this;
}

```

Spartos Testas su 1M-10M ra

[extra_cpp/testavimas.cpp](#) integravo [Vector](#) test:

```

// Trij strategij spartos palyginimas
Vector<Studentas> studentai;
studentai.reserve(1000000);

// Strategija 1: Dvigube konteineri manipuliacijos
// Strategija 2: Vienas konteineris su trimu i pagrindo
// Strategija 3: In-place riavimas ir dalijimas

// Visas ciklas beveik 2x greitesnis su reserve() optimizavimu

```

Google Test: [Vector<T>](#) Test Suite (v3.0 nauja!)

Visas [Vector](#) API testuojamas per Google Test framework - 27 testai, visi praeina Test List:

#	Testas	Apraas
1	VectorTest.PushBack	push_back() operacija
2	VectorTest.PopBack	pop_back() operacija
3	VectorTest.Size	size() funkcija
4	VectorTest.Empty	empty() check
5	VectorTest.Access	operator[] ir at()
6	VectorTest.FrontBack	front() ir back()
7	VectorTest.Reserve	reserve() atmintis
8	VectorTest.Clear	clear() operacija
9	VectorTest.ShrinkToFit	shrink_to_fit()
10	VectorTest.Resize	resize() metody
11	VectorTest.Swap	swap() metodas
12	VectorTest.ComparisonOperators	== ir != operatoriai
13	VectorTest.LessGreaterOperators	< ir > operatoriai
14	VectorTest.BeginEndIterators	begin()/end() iteratoriai
15	VectorTest.Assign	assign() metodas
16	VectorTest.Insert	insert() operacija
17	VectorTest.Erase	erase() operacija

#	Testas	Apraas
18	VectorTest.EmplaceBack	emplace_back()
19	VectorTest.Emplace	emplace() vietos konstravimas
20	VectorTest.MaxSize	max_size()
21	VectorTest.Data	data() raw pointer
22	VectorTest.CopyConstructor	Copy constructor
23	VectorTest.MoveConstructor	Move constructor
24	VectorTest.CopyAssignment	Copy assignment
25	VectorTest.MoveAssignment	Move assignment
26	VectorTest.ReverseIterators	rbegin()/rend()
27	VectorTest.ConversionFromStdVector	std::vector konversija

Paleidimas:

```
# Linux/macOS
cd build
./test_programa
```

```
# Windows
cd build
.\Release\test_programa.exe
```

```
# Atrinktas test
./test_programa --gtest_filter=VectorTest.PushBack
```

Rezultatas:

```
[=====] Running 27 tests from VectorTest
[ PASSED ] 27 tests
```

v2.0 relisas: Unit Testai (Google Test) ir Doxygen Dokumentacija

Unit Testai (v2.0)

Projektas naudoja hibridin test strategij:

- 13 test (originals i v1.5): assert() framework, Rule of Five ir I/O operacijos
- 5 testai (nauji v2.0): Google Test (gtest) framework, Rule of Five semantika 7

Rule of Five Metodai - assert() Framework (v1.5 pagrindiniai):

Testas	Metodas	Framework	Apraas
1	Default konstruktorius	assert()	Numatytosios reikms (A, BB, 0, 0.0)
2	Parametrizuotas konstruktorius	assert()	Duomen inicializavimas
3	Copy konstruktorius (Rule of Five #1)	assert()	Deep copy operacija
4	Copy assignment (Rule of Five #2)	assert()	Duomen kopijavimas ir self-assignment
5	Move konstruktorius (Rule of Five #3)	assert()	Itekli permimas, altinis tuias
6	Move assignment (Rule of Five #4)	assert()	Laikino objekto itekli permimas
7	Destruktorius (Rule of Five #5)	assert()	Tinkamas itekli atlaisvinimas (RAII)
8	I/O Round-trip	assert()	Ivestis ir vedimas (operator<< / operator>>)

Testas	Metodas	Framework	Apraas
9-10	Getters/Setters, vektoriaus ops	assert()	Duomen prieiga ir paymi operacijos

Papildomi Google Test Testai (v2.0):

Testas	Pavadinimas	Framework	Apraas
14	StudentasVectorTest.PushBackCopy Semantics	Google Test	Copy semantika su vektoriumi
15	StudentasVectorTest.PushBackMove Semantics	Google Test	Move semantika su vektoriumi
16	StudentasDeepCopyTest.Independent Copies	Google Test	Deep copy nepriklausomumas
17	StudentasSelfAssignmentTest.CopySafety Check	Google Test	Self-assignment saugumas
18	StudentasOperatorTest.Assignment Chaining	Google Test	Assignment operator chaining

Test Vykdymas

Automatinis vykdymas (rekomenduojama):

```
# Linux/macOS
./build.sh test
```

```
# Windows
powershell -ExecutionPolicy Bypass -File .\diegimas.ps1 -Stage "test"
```

Tiesiogias vykdymas:

```
# Linux/macOS
cd build
./test_programa
```

```
# Windows
cd out\build\x64-Debug
.\test_programa.exe
```

Tiktini Rezultatai:

```
=====
STUDENTAS KLASIS VISOS METODU TESTAI (v1.5)
+ GOOGLE TEST PAPILDYMAS (v2.0)
=====

[PRIIMTAS] Default konstruktorius testai...
...
Priimti testai: 50
Nepriimti testai: 0

VISI TESTAI PRIIMTI!

=====
Vykdome GOOGLE TEST testai...
=====

[=====] Running 5 tests from 4 test suites.
[ PASSED ] 5 tests.
```

Doxygen Dokumentacija (v2.0 nauja!)

Visa dokumentacija sugeneruojama i source kodo Doxygen komentar.

Sugeneruoti/atnaujinti dokumentacij:

```
# Automatinis bdas (rekomenduojama)
./build.sh docs # Linux/macOS
diegimas.ps1 -Stage docs # Windows
```

```
# Arba tiesiogias doxygen kvietimas
doxygen Doxyfile
```

Dokumentacijos Rezultatai:

- dokumentacija/html/index.html - Interaktyvi HTML dokumentacija (narom nuorodose)
- dokumentacija/latex/refman.pdf - Kompiliuota PDF dokumentacija (52 psl.)

Atidarytite dokumentacij:

- HTML: Atidaryti dokumentacija/html/index.html naryklje
- PDF: Atidaryti dokumentacija/latex/refman.pdf PDF skaitykloje

Diegimo Instrukcija

Sistemos Reikalavimai

Reikalavimas	Versija	Pastaba
CMake	3.10	Build sistema
C++ Kompiliatorius	C++17	MSVC / GCC / Clang
Google Test	v1.14.0	Automatikai parsiniama
Doxygen	naujausias	Dokumentacijai (parsisiuntimas: https://www.doxygen.nl/download.html)
TeX Live / LaTeX	2024+	PDF generavimui (parsisiuntimas: https://www.tug.org/texlive/)

1. Klonuojame repozicij

```
git clone https://github.com/ditassimoliunas-prog/objektuzdv1.git
cd objektuzdv1
git checkout v2.0
```

2. Automatinis diegimas (rekomenduojama)

Linux/macOS (su bash):

```
chmod +x build.sh
./build.sh all
```

Windows (su PowerShell):

```
Set-ExecutionPolicy -ExecutionPolicy Bypass -Scope Process
powershell -File .\diegimas.ps1 -Stage all
```

macOS (su Makefile):

```
make all
```

3. Rankinis diegimas ingsniai

Schritt 2a: Konfigruojame CMake

Linux/macOS:

```
mkdir build
cd build
cmake -DCMAKE_BUILD_TYPE=Release ..
```

Windows (MSVC/Ninja):

```
mkdir build
cd build
cmake -G "Visual Studio 17 2022" -A x64 ..
```

Schritt 2b: Kompiliuojame

Linux/macOS:

```
cmake --build . --config Release
```

Windows:

```
cmake --build . --config Release
```


Schritt 2c: Vykdomė testus

```
# Linux/macOS
./test_programa

# Windows
.\Release\test_programa.exe
```

Schritt 2d: Sugeneruojame dokumentacij

```
cd ..
doxygen Doxyfile
```

4. Patikrinimas

Po diegimo turtumte matyti:

- build/ direktorijoje kompiliuoti failai
- test_programa ivykdomas su 55 praejetais testais (50 assert + 5 gtest)
- dokumentacija/html/index.html - atidaryti naryklje
- dokumentacija/latex/refman.pdf - PDF dokumentacija

Naudojimosi Instrukcija

Pagrindins Programos Paleidimas

Linux/macOS:

```
./build/programa
```

Windows:

```
.\build\Release\programa.exe
```

Programos Meniu

Paleidus pagrindin program, galima rinktis i i opcij:

Opcij	Veikimas
1	Rankininis vedimas (vardas, pavard, 5 paymiai, egzaminas)
2	Generuoti tik paymius
3	Generuoti visus duomenis automatikai
4	Nuskaityti i failo
5	Sukurti testavimo failus
6	Atlikti spartos analiz
7	Baigti darb

Build Sumos Komandos

Linux/macOS (build.sh):

```
./build.sh build # Kompiliuoti
./build.sh test # Vykdyti testus
./build.sh docs # Sugeneruoti dokumentacij
./build.sh run # Paleisti program
./build.sh clean # Valyti build failus
./build.sh help # Pagalba
```

Windows (diegimas.ps1):

```
powershell -File .\diegimas.ps1 -Stage build # Kompiliuoti
powershell -File .\diegimas.ps1 -Stage test # Vykdyti testus
powershell -File .\diegimas.ps1 -Stage docs # Sugeneruoti dokumentacij
powershell -File .\diegimas.ps1 -Stage run # Paleisti program
powershell -File .\diegimas.ps1 -Stage clean # Valyti build failus
```

macOS (Makefile):

```
make build      # Kompiliuoti
make test      # Vykdyti testus
make docs      # Sugeneruoti dokumentacij
make run       # Paleisti program
make clean     # Valyti build failus
make info      # Pagalba
```

Projekto Fail Struktra

```
objektuzdv1/
CMakeLists.txt      # CMake build konfigracija (v2.0: Google Test FetchContent)
Doxyfile            # Doxygen konfigracija (v2.0: HTML + PDF)
Makefile            # Unix-style build helper (v2.0: build/test/docs/run/clean)
build.sh            # Linux/macOS build script (v2.0: automatinis diegimas)
diegimas.ps1        # Windows PowerShell script (v2.0: automatinis diegimas)
README.md           # is failas (v2.0: atnaujintas)
.gitignore          # Git taisykls (v2.0: dokumentacija jau versijuota)
header_files/
  Zmogus.h          # Klasi apraymai (interface)
  Studentas.h       # Abstrakti baz
  mat_funkcijos.h   # Studentas klas (Rule of Five)
  menu.h
  ...
extra_cpp/          # Implementacijos
  Zmogus.cpp         # Bazu klas implementacija
  Studentas.cpp      # Studentas klas implementacija
  mat_funkcijos.cpp
  menu.cpp
  ...
objekt1uzd.cpp       # Pagrindin programa (main)
test_studentas.cpp   # Unit testai (v2.0: 13x assert + 5x Google Test)
dokumentacija/      # Sugeneruota Doxygen dokumentacija (v2.0: versijuota!)
  html/
    index.html       # HTML dokumentacijos pradia
  latex/
    refman.tex
    refman.pdf       # PDF dokumentacija (52 psl.)
    ...
  ...
out/build/x64-Debug/ # Visual Studio build output (git ignored)
build/              # CMake build direktorija (git ignored)
  test_programa     # Kompiliuotas testas
  programa          # Kompiliuota pagrindin programa
  CMakeFiles/
.git/               # Git repositorija

---

## Klaid Sprendimas

### Klaida: "CMake not found"
bash
```

CMake instaliacija

Windows: <https://cmake.org/download/> (arba choco install cmake)

Linux: sudo apt install cmake

macOS: brew install cmake

```
### Klaida: "C++ compiler not found"
bash
```

Kompiliatoriaus instaliacija

Windows: Instaliuoti Visual Studio Build Tools arba Visual Studio Community

Linux: `sudo apt install build-essential g++`

macOS: `xcode-select --install`

```
### Klaida: "Doxygen not found"
bash
```

Doxygen parsisiuntimas ir instaliacija

<https://www.doxygen.nl/download.html>

macOS: `brew install doxygen`

Tada pridti doxygen PATH arba naudoti piln path:
`/usr/local/bin/doxygen`

```
### Klaida: "pdflatex not found"
bash
```

TeX Live parsisiuntimas (PDF generavimui)

Windows: <https://www.tug.org/texlive/> (instaliuoti texlive-full)

Linux: `sudo apt install texlive-latex-base texlive-latex-extra`

macOS: `brew install basicstex`

Po instaliavimo, pdflatex turt bti automatikai PATH'e

```
### Klaida: "Google Test linking errors"
bash
```

Paprastai isprendiama automatikai FetchContent'u CMake'e

Jei problema lieka:

1. Ivalykite build direktorij: `rm -rf build/`
2. I naujo sukonfigruokite: `cmake -B build -S .`
3. Perrankinkite: `cmake --build build --config Debug`

```
### Klaida: "test_programa.exe not found"
bash
```

Patikrinkite, ar build skminga:

```
cd build ls -la # Linux/macOS dir # Windows
```

Jei failas nra, perrankinkite: `cmake --build . --config Release`

...

DI Build Problem

Jei naudojate Visual Studio Code arba kit IDE ir kyla problemas:

1. Ivalykite CMake cache: `rm -rf build` (arba `rmdir /s build` Windows)
 2. I naujo sukonfigruokite projekt: `cmake -B build -S . -G "Visual Studio 17 2022"`
 3. Perrankinkite: `cmake --build build --config Release`
-

Release Istorija

v2.0 (Dabartinis)

- Integruotas Google Test framework (5 papildomi testai)
- Doxygen dokumentacija (HTML + PDF)
- CMake optimizuotas su FetchContent
- Automatiniai build scripti (build.sh, diegimas.ps1, Makefile)
- Visos Rule of Five metodai su `assert()` testais
- Dokumentacija versijuota git repository

v1.5

- Originals 13 `assert()` testai (Rule of Five + I/O)
 - Baz klas [Zmogus](#) su [Studentas](#) derivacija
 - Pagrindin meniu programa
-

Pagalba ir Suportas

Jei kyla klausim:

1. Patikrinkite dokumentacij: [dokumentacija/html/index.html](#)
 2. Perskaityti README sekcijas - greitasis startas aukiau
 3. Vykdyti testus - tai parodo, ar viskas veikia: `./test_programa`
 4. Kontaktuoti autori arba [Parayti Issue](#)
-

Autoriaus Informacija

Projekto Pavadinimas: objektuzdv1 (Objekcinio Programavimo Uduotis 1) Versija: v2.0 Sukurta: 2024-2026 Licenzija: MIT GitHub: <https://github.com/ditassimoliunas-prog/objektuzdv1>

Assert Rule of five testai:

Google.test testai:

skyrius 2

Hierarchijos Indeksas

2.1 Klasi hierarchija

is paveldjimo sraas yra beveik surikiuotas abcls tvarka:

deque< T >const_iterator	19
list< T >const_iterator	19
string::const_iterator	19
vector< T >const_iterator	19
deque< T >const_reverse_iterator	19
list< T >const_reverse_iterator	20
string::const_reverse_iterator	20
vector< T >const_reverse_iterator	20
deque< T >	20
exception	21
ifstream	21
invalid_argument	21
istream	21
deque< T >iterator	21
list< T >iterator	21
string::iterator	22
vector< T >iterator	22
list< T >	22
ofstream	22
ostream	22
deque< T >reverse_iterator	23
list< T >reverse_iterator	23
string::reverse_iterator	23
vector< T >reverse_iterator	23
runtime_error	23
string	23
stringstream	24
testing::Test	
VectorTest	42
Vector< T >	28
vector< T >	38
VectorIterator< T >	38
Zmogus	43
Studentas	24

skyrius 3

Klass Indeksas

3.1 Klass

Klass, struktros, sjungos ir ssajos su trumpais aprymais:

<code>deque< T >::const_iterator</code>	
STL iterator class	19
<code>list< T >::const_iterator</code>	
STL iterator class	19
<code>string::const_iterator</code>	
STL iterator class	19
<code>vector< T >::const_iterator</code>	
STL iterator class	19
<code>deque< T >::const_reverse_iterator</code>	
STL iterator class	19
<code>list< T >::const_reverse_iterator</code>	
STL iterator class	20
<code>string::const_reverse_iterator</code>	
STL iterator class	20
<code>vector< T >::const_reverse_iterator</code>	
STL iterator class	20
<code>deque< T ></code>	
STL class	20
<code>exception</code>	
STL class	21
<code>ifstream</code>	
STL class	21
<code>invalid_argument</code>	
STL class	21
<code>istream</code>	
STL class	21
<code>deque< T >::iterator</code>	
STL iterator class	21
<code>list< T >::iterator</code>	
STL iterator class	21
<code>string::iterator</code>	
STL iterator class	22
<code>vector< T >::iterator</code>	
STL iterator class	22
<code>list< T ></code>	
STL class	22
<code>ofstream</code>	
STL class	22
<code>ostream</code>	
STL class	22

deque< T >::reverse_iterator	
STL iterator class	23
list< T >::reverse_iterator	
STL iterator class	23
string::reverse_iterator	
STL iterator class	23
vector< T >::reverse_iterator	
STL iterator class	23
runtime_error	
STL class	23
string	
STL class	23
stringstream	
STL class	24
Studentas	24
Vector< T >	
Custom dynamic array container template	28
vector< T >	
STL class	38
VectorIterator< T >	
Random access iterator for Vector<T> container	38
VectorTest	42
Zmogus	
Abstrakti bazin klas bendram mogaus apraymui	43

skyrius 4

Failo Indeksas

4.1 Failai

Vis dokumentuot fail sraas su trumpais aprymais:

objekt1uzd.cpp	75
test_studentas.cpp	75
extra_cpp/duomenu_valdymas.cpp	47
extra_cpp/input.cpp	52
extra_cpp/mat_funkcijos.cpp	53
extra_cpp/menu.cpp	54
extra_cpp/output.cpp	56
extra_cpp/Studentas.cpp	57
extra_cpp/testavimas.cpp	58
extra_cpp/Zmogus.cpp	63
header_files/duomenu_valdymas.h	64
header_files/input.h	64
header_files/Mat_funkcijos.h	64
header_files/menu.h	64
header_files/output.h	64
header_files/Studentas.h	65
header_files/testavimas.h	66
header_files/Vector.h	66
header_files/Zmogus.h	74

skyrius 5

Klass Dokumentacija

5.1 deque< T >::const_iterator Klas

STL iterator class.

5.1.1 Smulkus apraymas

STL iterator class.

Dokumentacija iai klasei sugeneruota i i fail:

5.2 list< T >::const_iterator Klas

STL iterator class.

5.2.1 Smulkus apraymas

STL iterator class.

Dokumentacija iai klasei sugeneruota i i fail:

5.3 string::const_iterator Klas

STL iterator class.

5.3.1 Smulkus apraymas

STL iterator class.

Dokumentacija iai klasei sugeneruota i i fail:

5.4 vector< T >::const_iterator Klas

STL iterator class.

5.4.1 Smulkus apraymas

STL iterator class.

Dokumentacija iai klasei sugeneruota i i fail:

5.5 deque< T >::const_reverse_iterator Klas

STL iterator class.

5.5.1 Smulkus aprašymas

STL iterator class.

Dokumentacijaiai klasei sugeneruota i i fail:

5.6 `list< T >::const_reverse_iterator` Klas

STL iterator class.

5.6.1 Smulkus aprašymas

STL iterator class.

Dokumentacijaiai klasei sugeneruota i i fail:

5.7 `string::const_reverse_iterator` Klas

STL iterator class.

5.7.1 Smulkus aprašymas

STL iterator class.

Dokumentacijaiai klasei sugeneruota i i fail:

5.8 `vector< T >::const_reverse_iterator` Klas

STL iterator class.

5.8.1 Smulkus aprašymas

STL iterator class.

Dokumentacijaiai klasei sugeneruota i i fail:

5.9 `deque< T >` Klas ablonas

STL class.

Klasas

- class [iterator](#)
STL iterator class.
- class [const_iterator](#)
STL iterator class.
- class [reverse_iterator](#)
STL iterator class.
- class [const_reverse_iterator](#)
STL iterator class.

Vieji Atributai

- T elements
STL member.

5.9.1 Smulkus aprašymas

STL class.

Dokumentacijaiai klasei sugeneruota i i fail:

5.10 exception Klas

STL class.

5.10.1 Smulkus apraymas

STL class.

Dokumentacija iai klasei sugeneruota i i fail:

5.11 ifstream Klas

STL class.

5.11.1 Smulkus apraymas

STL class.

Dokumentacija iai klasei sugeneruota i i fail:

5.12 invalid_argument Klas

STL class.

5.12.1 Smulkus apraymas

STL class.

Dokumentacija iai klasei sugeneruota i i fail:

5.13 istream Klas

STL class.

5.13.1 Smulkus apraymas

STL class.

Dokumentacija iai klasei sugeneruota i i fail:

5.14 deque< T >::iterator Klas

STL iterator class.

5.14.1 Smulkus apraymas

STL iterator class.

Dokumentacija iai klasei sugeneruota i i fail:

5.15 list< T >::iterator Klas

STL iterator class.

5.15.1 Smulkus apraymas

STL iterator class.

Dokumentacija iai klasei sugeneruota i i fail:

5.16 string::iterator Klas

STL iterator class.

5.16.1 Smulkus apraymas

STL iterator class.

Dokumentacija iai klasei sugeneruota i i fail:

5.17 vector< T >::iterator Klas

STL iterator class.

5.17.1 Smulkus apraymas

STL iterator class.

Dokumentacija iai klasei sugeneruota i i fail:

5.18 list< T > Klas ablonas

STL class.

Klass

- class [iterator](#)
STL iterator class.
- class [const_iterator](#)
STL iterator class.
- class [reverse_iterator](#)
STL iterator class.
- class [const_reverse_iterator](#)
STL iterator class.

Viei Atributai

- T elements
STL member.

5.18.1 Smulkus apraymas

STL class.

Dokumentacija iai klasei sugeneruota i i fail:

5.19 ofstream Klas

STL class.

5.19.1 Smulkus apraymas

STL class.

Dokumentacija iai klasei sugeneruota i i fail:

5.20 ostream Klas

STL class.

5.20.1 Smulkus apraymas

STL class.

Dokumentacija iai klasei sugeneruota i i fail:

5.21 deque< T >::reverse_iterator Klas

STL iterator class.

5.21.1 Smulkus apraymas

STL iterator class.

Dokumentacija iai klasei sugeneruota i i fail:

5.22 list< T >::reverse_iterator Klas

STL iterator class.

5.22.1 Smulkus apraymas

STL iterator class.

Dokumentacija iai klasei sugeneruota i i fail:

5.23 string::reverse_iterator Klas

STL iterator class.

5.23.1 Smulkus apraymas

STL iterator class.

Dokumentacija iai klasei sugeneruota i i fail:

5.24 vector< T >::reverse_iterator Klas

STL iterator class.

5.24.1 Smulkus apraymas

STL iterator class.

Dokumentacija iai klasei sugeneruota i i fail:

5.25 runtime_error Klas

STL class.

5.25.1 Smulkus apraymas

STL class.

Dokumentacija iai klasei sugeneruota i i fail:

5.26 string Klas

STL class.

Klass

- class `iterator`
STL iterator class.
- class `const_iterator`
STL iterator class.
- class `reverse_iterator`
STL iterator class.
- class `const_reverse_iterator`
STL iterator class.

5.26.1 Smulkus apraymas

STL class.

Dokumentacija iai klasei sugeneruota i i fail:

5.27 stringstream Klas

STL class.

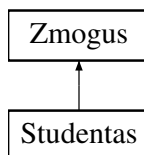
5.27.1 Smulkus apraymas

STL class.

Dokumentacija iai klasei sugeneruota i i fail:

5.28 Studentas Klas

Paveldimumo diagrama Studentas:



Viei Metodai

- `Studentas` (`string` v, `string` p, `int` e)
- `Studentas` (`std::istream` &is)
- `Studentas` (`const Studentas` &other)
- `Studentas` & `operator=` (`const Studentas` &other)
- `Studentas` (`Studentas` &&other) `noexcept`
- `Studentas` & `operator=` (`Studentas` &&other) `noexcept`
- `const Vector< int >` & `getPaz` () `const`
- `Vector< int >` & `getPaz` ()
- `int` `getEgz` () `const`
- `double` `getRez` () `const`
- `double` `getMed` () `const`
- `void` `setEgz` (`int` e)
- `void` `setRez` (`double` r)
- `void` `setMed` (`double` m)
- `void` `addPaz` (`int` p)
- `void` `clearPaz` ()
- `void` `reservePaz` (`size_t` n)
- `bool` `isPazEmpty` () `const`

- void `setPaz` (const `Vector< int > &p`)
- `std::istream & readStudent` (`std::istream &is`)
- virtual `string getInfo` () const override

Grina mogaus informacij is metodos turi bti implementuotas visose ivestinsse klasse.

Viei Metodai inherited from `Zmogus`

- `Zmogus` (const `string &v`, const `string &p`)
Parametrizuotas konstruktorius.
- `Zmogus` ()
Parametrizuotas konstruktorius su numatytosiomis reikmmis.
- virtual `Zmogus` ()
Virtualus destruktorius - reikalingas polimorfizmui.
- `string getVardas` () const
- `string getPavarde` () const
- void `setVardas` (const `string &v`)
Nustato mogaus vard
- void `setPavarde` (const `string &p`)
Nustato mogaus pavard

Draugai

- `std::ostream & operator<<` (`std::ostream &os`, const `Studentas &s`)
- `std::istream & operator>>` (`std::istream &is`, `Studentas &s`)

Additional Inherited Members

Apsaugoti Atributai inherited from `Zmogus`

- `string vardas_`
- `string pavarde_`

5.28.1 Smulkus apraymas

Apibrimas failo `Studentas.h` eilutje 14.

5.28.2 Konstruktoriaus ir Destruktoriaus Dokumentacija

5.28.2.1 Studentas() [1/5]

`Studentas::Studentas ()` [inline]

Apibrimas failo `Studentas.h` eilutje 29.

5.28.2.2 Studentas() [2/5]

`Studentas::Studentas (`
`string v,`
`string p,`
`int e)` [inline]

Apibrimas failo `Studentas.h` eilutje 32.

5.28.2.3 Studentas() [3/5]

`Studentas::Studentas (`
`std::istream & is)`

Apibrimas failo `Studentas.cpp` eilutje 8.

5.28.2.4 Studentas()

Studentas::Studentas ()

Apibrimas failo [Studentas.cpp](#) eilutje 15.

5.28.2.5 Studentas() [4/5]

Studentas::Studentas (

const [Studentas](#) & other)

Apibrimas failo [Studentas.cpp](#) eilutje 20.

5.28.2.6 Studentas() [5/5]

Studentas::Studentas (

[Studentas](#) && other) [noexcept]

Apibrimas failo [Studentas.cpp](#) eilutje 39.

5.28.3 Metod Dokumentacija

5.28.3.1 addPaz()

void Studentas::addPaz (

int p) [inline]

Apibrimas failo [Studentas.h](#) eilutje 67.

5.28.3.2 clearPaz()

void Studentas::clearPaz () [inline]

Apibrimas failo [Studentas.h](#) eilutje 68.

5.28.3.3 getEgz()

int Studentas::getEgz () const [inline]

Apibrimas failo [Studentas.h](#) eilutje 57.

5.28.3.4 getInfo()

[string](#) Studentas::getInfo () const [override], [virtual]

Grina mogaus informacij is metodos turi bti implementuotas visose ivestinese klasse.

Realizuoja [Zmogus](#).

Apibrimas failo [Studentas.cpp](#) eilutje 68.

Ryiai [Zmogus::getPavarde\(\)](#), ir [Zmogus::getVardas\(\)](#).

5.28.3.5 getMed()

double Studentas::getMed () const [inline]

Apibrimas failo [Studentas.h](#) eilutje 59.

5.28.3.6 getPaz() [1/2]

[Vector](#)< int > & Studentas::getPaz () [inline]

Apibrimas failo [Studentas.h](#) eilutje 56.

5.28.3.7 getPaz() [2/2]

const [Vector](#)< int > & Studentas::getPaz () const [inline]

Apibrimas failo [Studentas.h](#) eilutje 55.

5.28.3.8 getRez()

double Studentas::getRez () const [inline]

Apibrimas failo [Studentas.h](#) eilutje 58.

5.28.3.9 isPazEmpty()

bool Studentas::isPazEmpty () const [inline]
Apibrimas failo [Studentas.h](#) eilutje 70.

5.28.3.10 operator=() [1/2]

[Studentas](#) & Studentas::operator= (
const [Studentas](#) & other)
Apibrimas failo [Studentas.cpp](#) eilutje 26.

5.28.3.11 operator=() [2/2]

[Studentas](#) & Studentas::operator= (
[Studentas](#) && other) [noexcept]
Apibrimas failo [Studentas.cpp](#) eilutje 50.

5.28.3.12 readStudent()

std::istream & Studentas::readStudent (
std::istream & is)
Apibrimas failo [Studentas.cpp](#) eilutje 91.

5.28.3.13 reservePaz()

void Studentas::reservePaz (
size_t n) [inline]
Apibrimas failo [Studentas.h](#) eilutje 69.

5.28.3.14 setEgz()

void Studentas::setEgz (
int e) [inline]
Apibrimas failo [Studentas.h](#) eilutje 62.

5.28.3.15 setMed()

void Studentas::setMed (
double m) [inline]
Apibrimas failo [Studentas.h](#) eilutje 64.

5.28.3.16 setPaz()

void Studentas::setPaz (
const [Vector](#)< int > & p) [inline]
Apibrimas failo [Studentas.h](#) eilutje 71.

5.28.3.17 setRez()

void Studentas::setRez (
double r) [inline]
Apibrimas failo [Studentas.h](#) eilutje 63.

5.28.4 Draugik Ir Susijusi Funkcij Dokumentacija

5.28.4.1 operator<<

std::ostream & operator<< (
std::ostream & os,
const [Studentas](#) & s) [friend]
Apibrimas failo [Studentas.cpp](#) eilutje 111.

5.28.4.2 operator>>

```
std::istream & operator>> (
    std::istream & is,
    Studentas & s) [friend]
```

Apibrimas failo [Studentas.cpp](#) eilutje 128.

Dokumentacija iai klasei sugeneruota i i fail:

- header_files/Studentas.h
- extra_cpp/Studentas.cpp

5.29 Vector< T > Klas ablonas

Custom dynamic array container template.

```
#include <Vector.h>
```

Viei Tipai

- using [iterator](#) = [VectorIterator](#)<T>
- using [const_iterator](#) = [VectorIterator](#)<const T>
- using [reverse_iterator](#) = std::reverse_iterator<iterator>
- using [const_reverse_iterator](#) = std::reverse_iterator<const_iterator>

Viei Metodai

- [Vector](#) (size_t capacity)
- [Vector](#) (size_t size, const T &value)
- [Vector](#) (const std::vector< T > &other)
- [Vector](#) (const [Vector](#) &other)
- [Vector](#) & operator= (const [Vector](#) &other)
- [Vector](#) ([Vector](#) &&other) noexcept
- [Vector](#) & operator= ([Vector](#) &&other) noexcept
- size_t [size](#) () const
- size_t [capacity](#) () const
- bool [empty](#) () const
- T & [operator\[\]](#) (size_t index)
- const T & [operator\[\]](#) (size_t index) const
- T & [at](#) (size_t index)
- const T & [at](#) (size_t index) const
- void [clear](#) ()
- T & [front](#) ()
- const T & [front](#) () const
- T & [back](#) ()
- const T & [back](#) () const
- void [reserve](#) (size_t new_capacity)
- void [shrink_to_fit](#) ()
- void [resize](#) (size_t new_size)
- void [resize](#) (size_t new_size, const T &value)
- void [swap](#) ([Vector](#) &other) noexcept
- bool [operator==](#) (const [Vector](#) &other) const
- bool [operator!=](#) (const [Vector](#) &other) const
- bool [operator<](#) (const [Vector](#) &other) const
- bool [operator>](#) (const [Vector](#) &other) const
- bool [operator<=](#) (const [Vector](#) &other) const
- bool [operator>=](#) (const [Vector](#) &other) const
- iterator [begin](#) ()
- iterator [end](#) ()

- `const_iterator` `begin` () const
- `const_iterator` `end` () const
- `const_iterator` `cbegin` () const
- `const_iterator` `cend` () const
- `reverse_iterator` `rbegin` ()
- `reverse_iterator` `rend` ()
- `const_reverse_iterator` `rbegin` () const
- `const_reverse_iterator` `rend` () const
- `const_reverse_iterator` `crbegin` () const
- `const_reverse_iterator` `crend` () const
- `void` `assign` (size_t count, const T &value)
- `template<typename InputIt>`
`void` `assign` (InputIt first, InputIt last)
- `iterator` `insert` (const_iterator pos, const T &value)
- `iterator` `insert` (iterator pos, const T &value)
- `template<typename InputIt>`
`iterator` `insert` (const_iterator pos, InputIt first, InputIt last)
- `template<typename InputIt>`
`iterator` `insert` (iterator pos, InputIt first, InputIt last)
- `iterator` `erase` (const_iterator pos)
- `iterator` `erase` (const_iterator first, const_iterator last)
- `iterator` `erase` (iterator pos)
- `iterator` `erase` (iterator first, iterator last)
- `template<typename... Args>`
`void` `emplace_back` (Args &&... args)
- `template<typename... Args>`
`iterator` `emplace` (const_iterator pos, Args &&... args)
- `size_t` `max_size` () const
- `T` `data` ()
- `const T` `data` () const
- `size_t` `get_reallocation_count` () const
- `void` `reset_reallocation_count` ()
- `void` `push_back` (const T &value)
- `void` `push_back` (T &&value)
- `void` `pop_back` ()

5.29.1 Smulkus apraymas

```
template<typename T>
class Vector< T >
```

Custom dynamic array container template.

`Vector<T>` is a generic container that stores elements of type T in a dynamically allocated array. It provides O(1) amortized time complexity for `push_back()`, while maintaining contiguous memory layout like `std::vector`.

The class implements the Rule of Five pattern and fully supports deep copy and move semantics.

Key Features:

- Dynamic memory management (automatic resizing)
- Random access iterators (`begin()`, `end()`, `rbegin()`, `rend()`)
- Copy and move semantics (Rule of Five)
- Standard container operations (`push_back`, `pop_back`, `insert`, `erase`, etc.)
- Comparable performance to `std::vector` for most operations

Memory Strategy:

- When capacity is exhausted, reallocates with roughly 1.5x growth factor
- Tracks reallocation count for performance analysis
- Supports manual reserve() for optimization

Example Usage:

```
Vector<int> v;
v.push_back(10);
v.push_back(20);
v.reserve(100);    // Pre-allocate capacity

for (int elem : v) {
    std::cout << elem << " ";
}
```

Template Parameters

T	Element type (must support copy and move semantics)
---	---

Taip pat irti

[VectorIterator](#)

`std::vector`

Pavyzdiai

C:/Users/ditas/source/repos/ditassimoliunas-prog/objektuzdv3/header_files/Vector.h.

Apibrimas failo [Vector.h](#) eilutje 132.

5.29.2 Tipo Apraymo Dokumentacija

5.29.2.1 const_iterator

```
template<typename T>
using Vector< T >::const_iterator = VectorIterator<const T>
Apibrimas failo Vector.h eilutje 339.
```

5.29.2.2 const_reverse_iterator

```
template<typename T>
using Vector< T >::const_reverse_iterator = std::reverse_iterator<const_iterator>
Apibrimas failo Vector.h eilutje 351.
```

5.29.2.3 iterator

```
template<typename T>
using Vector< T >::iterator = VectorIterator<T>
Apibrimas failo Vector.h eilutje 338.
```

5.29.2.4 reverse_iterator

```
template<typename T>
using Vector< T >::reverse_iterator = std::reverse_iterator<iterator>
Apibrimas failo Vector.h eilutje 350.
```

5.29.3 Konstruktoriaus ir Destruktoriaus Dokumentacija

5.29.3.1 Vector() [1/6]

```
template<typename T>
Vector< T >::Vector () [inline]
Apibrimas failo Vector.h eilutje 154.
```

5.29.3.2 Vector() [2/6]

```
template<typename T>
Vector< T >::Vector (
    size_t capacity) [inline], [explicit]
```

Apibrimas failo [Vector.h](#) eilutje 155.

5.29.3.3 Vector() [3/6]

```
template<typename T>
Vector< T >::Vector (
    size_t size,
    const T & value) [inline]
```

Apibrimas failo [Vector.h](#) eilutje 158.

5.29.3.4 Vector() [4/6]

```
template<typename T>
Vector< T >::Vector (
    const std::vector< T > & other) [inline]
```

Apibrimas failo [Vector.h](#) eilutje 164.

5.29.3.5 Vector()

```
template<typename T>
Vector< T >::Vector () [inline]
```

Apibrimas failo [Vector.h](#) eilutje 169.

5.29.3.6 Vector() [5/6]

```
template<typename T>
Vector< T >::Vector (
    const Vector< T > & other) [inline]
```

Apibrimas failo [Vector.h](#) eilutje 171.

5.29.3.7 Vector() [6/6]

```
template<typename T>
Vector< T >::Vector (
    Vector< T > && other) [inline], [noexcept]
```

Apibrimas failo [Vector.h](#) eilutje 187.

5.29.4 Metod Dokumentacija

5.29.4.1 assign() [1/2]

```
template<typename T>
template<typename InputIt>
void Vector< T >::assign (
    InputIt first,
    InputIt last) [inline]
```

Apibrimas failo [Vector.h](#) eilutje 374.

5.29.4.2 assign() [2/2]

```
template<typename T>
void Vector< T >::assign (
    size_t count,
    const T & value) [inline]
```

Apibrimas failo [Vector.h](#) eilutje 362.

5.29.4.3 at() [1/2]

```
template<typename T>
T & Vector< T >::at (
    size_t index) [inline]
```

Apibrimas failo [Vector.h](#) eilutje 216.

5.29.4.4 at() [2/2]

```
template<typename T>
const T & Vector< T >::at (
    size_t index) const [inline]
```

Apibrimas failo [Vector.h](#) eilutje 223.

5.29.4.5 back() [1/2]

```
template<typename T>
T & Vector< T >::back () [inline]
```

Apibrimas failo [Vector.h](#) eilutje 248.

5.29.4.6 back() [2/2]

```
template<typename T>
const T & Vector< T >::back () const [inline]
```

Apibrimas failo [Vector.h](#) eilutje 255.

5.29.4.7 begin() [1/2]

```
template<typename T>
iterator Vector< T >::begin () [inline]
```

Apibrimas failo [Vector.h](#) eilutje 341.

5.29.4.8 begin() [2/2]

```
template<typename T>
const_iterator Vector< T >::begin () const [inline]
```

Apibrimas failo [Vector.h](#) eilutje 344.

5.29.4.9 capacity()

```
template<typename T>
size_t Vector< T >::capacity () const [inline]
```

Apibrimas failo [Vector.h](#) eilutje 210.

5.29.4.10 cbegin()

```
template<typename T>
const_iterator Vector< T >::cbegin () const [inline]
```

Apibrimas failo [Vector.h](#) eilutje 347.

5.29.4.11 cend()

```
template<typename T>
const_iterator Vector< T >::cend () const [inline]
```

Apibrimas failo [Vector.h](#) eilutje 348.

5.29.4.12 clear()

```
template<typename T>
void Vector< T >::clear () [inline]
```

Apibrimas failo [Vector.h](#) eilutje 230.

5.29.4.13 crbegin()

```
template<typename T>
const_reverse_iterator Vector< T >::crbegin () const [inline]
Apibrimas failo Vector.h eilutje 359.
```

5.29.4.14 crend()

```
template<typename T>
const_reverse_iterator Vector< T >::crend () const [inline]
Apibrimas failo Vector.h eilutje 360.
```

5.29.4.15 data() [1/2]

```
template<typename T>
T Vector< T >::data () [inline]
Apibrimas failo Vector.h eilutje 588.
```

5.29.4.16 data() [2/2]

```
template<typename T>
const T Vector< T >::data () const [inline]
Apibrimas failo Vector.h eilutje 589.
```

5.29.4.17 emplace()

```
template<typename T>
template<typename... Args>
iterator Vector< T >::emplace (
    const_iterator pos,
    Args &&... args) [inline]
Apibrimas failo Vector.h eilutje 564.
```

5.29.4.18 emplace_back()

```
template<typename T>
template<typename... Args>
void Vector< T >::emplace_back (
    Args &&... args) [inline]
Apibrimas failo Vector.h eilutje 554.
```

5.29.4.19 empty()

```
template<typename T>
bool Vector< T >::empty () const [inline]
Apibrimas failo Vector.h eilutje 211.
```

5.29.4.20 end() [1/2]

```
template<typename T>
iterator Vector< T >::end () [inline]
Apibrimas failo Vector.h eilutje 342.
```

5.29.4.21 end() [2/2]

```
template<typename T>
const_iterator Vector< T >::end () const [inline]
Apibrimas failo Vector.h eilutje 345.
```

5.29.4.22 erase() [1/4]

```
template<typename T>
iterator Vector< T >::erase (
    const_iterator first,
    const_iterator last) [inline]
```

Apibrimas failo [Vector.h](#) eilutje 503.

5.29.4.23 erase() [2/4]

```
template<typename T>
iterator Vector< T >::erase (
    const_iterator pos) [inline]
```

Apibrimas failo [Vector.h](#) eilutje 489.

5.29.4.24 erase() [3/4]

```
template<typename T>
iterator Vector< T >::erase (
    iterator first,
    iterator last) [inline]
```

Apibrimas failo [Vector.h](#) eilutje 536.

5.29.4.25 erase() [4/4]

```
template<typename T>
iterator Vector< T >::erase (
    iterator pos) [inline]
```

Apibrimas failo [Vector.h](#) eilutje 521.

5.29.4.26 front() [1/2]

```
template<typename T>
T & Vector< T >::front () [inline]
```

Apibrimas failo [Vector.h](#) eilutje 234.

5.29.4.27 front() [2/2]

```
template<typename T>
const T & Vector< T >::front () const [inline]
```

Apibrimas failo [Vector.h](#) eilutje 241.

5.29.4.28 get_reallocation_count()

```
template<typename T>
size_t Vector< T >::get_reallocation_count () const [inline]
```

Apibrimas failo [Vector.h](#) eilutje 591.

5.29.4.29 insert() [1/4]

```
template<typename T>
iterator Vector< T >::insert (
    const_iterator pos,
    const T & value) [inline]
```

Apibrimas failo [Vector.h](#) eilutje 380.

5.29.4.30 insert() [2/4]

```
template<typename T>
template<typename InputIt>
iterator Vector< T >::insert (
```

```

    const_iterator pos,
    InputIt first,
    InputIt last) [inline]

```

Apibrimas failo [Vector.h](#) eilutje 425.

5.29.4.31 insert() [3/4]

```

template<typename T>
iterator Vector< T >::insert (
    iterator pos,
    const T & value) [inline]

```

Apibrimas failo [Vector.h](#) eilutje 401.

5.29.4.32 insert() [4/4]

```

template<typename T>
template<typename InputIt>
iterator Vector< T >::insert (
    iterator pos,
    InputIt first,
    InputIt last) [inline]

```

Apibrimas failo [Vector.h](#) eilutje 458.

5.29.4.33 max_size()

```

template<typename T>
size_t Vector< T >::max_size () const [inline]

```

Apibrimas failo [Vector.h](#) eilutje 584.

5.29.4.34 operator"!="()

```

template<typename T>
bool Vector< T >::operator!= (
    const Vector< T > & other) const [inline]

```

Apibrimas failo [Vector.h](#) eilutje 314.

5.29.4.35 operator<()

```

template<typename T>
bool Vector< T >::operator< (
    const Vector< T > & other) const [inline]

```

Apibrimas failo [Vector.h](#) eilutje 318.

5.29.4.36 operator<=()

```

template<typename T>
bool Vector< T >::operator<= (
    const Vector< T > & other) const [inline]

```

Apibrimas failo [Vector.h](#) eilutje 330.

5.29.4.37 operator=() [1/2]

```

template<typename T>
Vector & Vector< T >::operator= (
    const Vector< T > & other) [inline]

```

Apibrimas failo [Vector.h](#) eilutje 175.

5.29.4.38 operator=() [2/2]

```
template<typename T>
Vector & Vector< T >::operator= (
    Vector< T > && other) [inline], [noexcept]
```

Apibrimas failo [Vector.h](#) eilutje 194.

5.29.4.39 operator==()

```
template<typename T>
bool Vector< T >::operator== (
    const Vector< T > & other) const [inline]
```

Apibrimas failo [Vector.h](#) eilutje 306.

5.29.4.40 operator>()

```
template<typename T>
bool Vector< T >::operator> (
    const Vector< T > & other) const [inline]
```

Apibrimas failo [Vector.h](#) eilutje 326.

5.29.4.41 operator>=()

```
template<typename T>
bool Vector< T >::operator>= (
    const Vector< T > & other) const [inline]
```

Apibrimas failo [Vector.h](#) eilutje 334.

5.29.4.42 operator[]() [1/2]

```
template<typename T>
T & Vector< T >::operator[] (
    size_t index) [inline]
```

Apibrimas failo [Vector.h](#) eilutje 213.

5.29.4.43 operator[]() [2/2]

```
template<typename T>
const T & Vector< T >::operator[] (
    size_t index) const [inline]
```

Apibrimas failo [Vector.h](#) eilutje 214.

5.29.4.44 pop_back()

```
template<typename T>
void Vector< T >::pop_back () [inline]
```

Apibrimas failo [Vector.h](#) eilutje 672.

5.29.4.45 push_back() [1/2]

```
template<typename T>
void Vector< T >::push_back (
    const T & value) [inline]
```

Pavyzdiai

[C:/Users/ditas/source/repos/ditassimoliunas-prog/objektuzdv3/header_files/Vector.h](#).

Apibrimas failo [Vector.h](#) eilutje 615.

5.29.4.46 push_back() [2/2]

```
template<typename T>
void Vector< T >::push_back (
    T && value) [inline]
```

Apibrimas failo [Vector.h](#) eilutje 644.

5.29.4.47 rbegin() [1/2]

```
template<typename T>
reverse_iterator Vector< T >::rbegin () [inline]
```

Apibrimas failo [Vector.h](#) eilutje 353.

5.29.4.48 rbegin() [2/2]

```
template<typename T>
const_reverse_iterator Vector< T >::rbegin () const [inline]
```

Apibrimas failo [Vector.h](#) eilutje 356.

5.29.4.49 rend() [1/2]

```
template<typename T>
reverse_iterator Vector< T >::rend () [inline]
```

Apibrimas failo [Vector.h](#) eilutje 354.

5.29.4.50 rend() [2/2]

```
template<typename T>
const_reverse_iterator Vector< T >::rend () const [inline]
```

Apibrimas failo [Vector.h](#) eilutje 357.

5.29.4.51 reserve()

```
template<typename T>
void Vector< T >::reserve (
    size_t new_capacity) [inline]
```

Apibrimas failo [Vector.h](#) eilutje 262.

5.29.4.52 reset_reallocation_count()

```
template<typename T>
void Vector< T >::reset_reallocation_count () [inline]
```

Apibrimas failo [Vector.h](#) eilutje 592.

5.29.4.53 resize() [1/2]

```
template<typename T>
void Vector< T >::resize (
    size_t new_size) [inline]
```

Apibrimas failo [Vector.h](#) eilutje 280.

5.29.4.54 resize() [2/2]

```
template<typename T>
void Vector< T >::resize (
    size_t new_size,
    const T & value) [inline]
```

Apibrimas failo [Vector.h](#) eilutje 287.

5.29.4.55 shrink_to_fit()

```
template<typename T>
void Vector< T >::shrink_to_fit () [inline]
Apibrimas failo Vector.h eilutje 268.
```

5.29.4.56 size()

```
template<typename T>
size_t Vector< T >::size () const [inline]
Apibrimas failo Vector.h eilutje 209.
```

5.29.4.57 swap()

```
template<typename T>
void Vector< T >::swap (
    Vector< T > & other) [inline], [noexcept]
```

Apibrimas failo [Vector.h](#) eilutje 299.

Dokumentacija iai klasei sugeneruota i io failo:

- `header_files/Vector.h`

5.30 vector< T > Klas ablonas

STL class.

Klass

- class [iterator](#)
STL iterator class.
- class [const_iterator](#)
STL iterator class.
- class [reverse_iterator](#)
STL iterator class.
- class [const_reverse_iterator](#)
STL iterator class.

Viei Atributai

- T elements
STL member.

5.30.1 Smulkus apraymas

STL class.

Dokumentacija iai klasei sugeneruota i i fail:

5.31 VectorIterator< T > Klas ablonas

Random access iterator for [Vector](#)<T> container.

```
#include <Vector.h>
```

Viei Tipai

- using [iterator_category](#) = std::random_access_iterator_tag
- using [value_type](#) = T
- using [difference_type](#) = std::ptrdiff_t
- using [pointer](#) = T
- using [reference](#) = T&

Vieji Metodai

- [VectorIterator](#) (T ptr=nullptr)
- reference [operator](#) () const
- pointer [operator->](#) () const
- [VectorIterator](#) & [operator++](#) ()
- [VectorIterator](#) [operator++](#) (int)
- [VectorIterator](#) & [operator--](#) ()
- [VectorIterator](#) [operator--](#) (int)
- [VectorIterator](#) [operator+](#) (difference_type n) const
- [VectorIterator](#) [operator-](#) (difference_type n) const
- [VectorIterator](#) & [operator+=](#) (difference_type n)
- [VectorIterator](#) & [operator-=](#) (difference_type n)
- reference [operator\[\]](#) (difference_type n) const
- bool [operator==](#) (const [VectorIterator](#) &other) const
- bool [operator!=](#) (const [VectorIterator](#) &other) const
- bool [operator<](#) (const [VectorIterator](#) &other) const
- bool [operator>](#) (const [VectorIterator](#) &other) const
- bool [operator<=](#) (const [VectorIterator](#) &other) const
- bool [operator>=](#) (const [VectorIterator](#) &other) const
- difference_type [operator-](#) (const [VectorIterator](#) &other) const

5.31.1 Smulkus aprašymas

```
template<typename T>
class VectorIterator< T >
```

Random access iterator for [Vector<T>](#) container.

[VectorIterator](#) provides bidirectional and random access to [Vector](#) elements. Supports all standard iterator operations including comparison, arithmetic, and dereferencing.

Template Parameters

T	The element type that the iterator points to
---	--

Taip pat irti

[Vector](#)

Pavyzdiai

[C:/Users/ditas/source/repos/ditassimoliunas-prog/objektuzdv3/header_files/Vector.h.](#)

Apibrimas failo [Vector.h](#) eilutje 22.

5.31.2 Tipo Aprašymo Dokumentacija

5.31.2.1 difference_type

```
template<typename T>
using VectorIterator< T >::difference_type = std::ptrdiff_t
```

Apibrimas failo [Vector.h](#) eilutje 26.

5.31.2.2 iterator_category

```
template<typename T>
using VectorIterator< T >::iterator_category = std::random_access_iterator_tag
```

Apibrimas failo [Vector.h](#) eilutje 24.

5.31.2.3 pointer

```
template<typename T>
using VectorIterator< T >::pointer = T
Apibrimas failo Vector.h eilutje 27.
```

5.31.2.4 reference

```
template<typename T>
using VectorIterator< T >::reference = T&
Apibrimas failo Vector.h eilutje 28.
```

5.31.2.5 value_type

```
template<typename T>
using VectorIterator< T >::value_type = T
Apibrimas failo Vector.h eilutje 25.
```

5.31.3 Konstruktoriaus ir Destruktoriaus Dokumentacija

5.31.3.1 VectorIterator()

```
template<typename T>
VectorIterator< T >::VectorIterator (
    T ptr = nullptr) [inline]
Apibrimas failo Vector.h eilutje 34.
```

5.31.4 Metod Dokumentacija

5.31.4.1 operator"!=()"

```
template<typename T>
bool VectorIterator< T >::operator!= (
    const VectorIterator< T > & other) const [inline]
Apibrimas failo Vector.h eilutje 80.
```

5.31.4.2 operator()

```
template<typename T>
reference VectorIterator< T >::operator () const [inline]
Apibrimas failo Vector.h eilutje 36.
```

5.31.4.3 operator+()

```
template<typename T>
VectorIterator VectorIterator< T >::operator+ (
    difference_type n) const [inline]
Apibrimas failo Vector.h eilutje 59.
```

5.31.4.4 operator++() [1/2]

```
template<typename T>
VectorIterator & VectorIterator< T >::operator++ () [inline]
Apibrimas failo Vector.h eilutje 39.
```

5.31.4.5 operator++() [2/2]

```
template<typename T>
VectorIterator VectorIterator< T >::operator++ (
    int ) [inline]
Apibrimas failo Vector.h eilutje 43.
```


5.31.4.6 operator+=()

```
template<typename T>
VectorIterator & VectorIterator< T >::operator+=(
    difference_type n) [inline]
```

Apibrimas failo [Vector.h](#) eilutje 66.

5.31.4.7 operator-() [1/2]

```
template<typename T>
difference_type VectorIterator< T >::operator- (
    const VectorIterator< T > & other) const [inline]
```

Apibrimas failo [Vector.h](#) eilutje 86.

5.31.4.8 operator-() [2/2]

```
template<typename T>
VectorIterator VectorIterator< T >::operator- (
    difference_type n) const [inline]
```

Apibrimas failo [Vector.h](#) eilutje 62.

5.31.4.9 operator--() [1/2]

```
template<typename T>
VectorIterator & VectorIterator< T >::operator-- () [inline]
```

Apibrimas failo [Vector.h](#) eilutje 49.

5.31.4.10 operator--() [2/2]

```
template<typename T>
VectorIterator VectorIterator< T >::operator-- (
    int ) [inline]
```

Apibrimas failo [Vector.h](#) eilutje 53.

5.31.4.11 operator==()

```
template<typename T>
VectorIterator & VectorIterator< T >::operator== (
    difference_type n) [inline]
```

Apibrimas failo [Vector.h](#) eilutje 70.

5.31.4.12 operator->()

```
template<typename T>
pointer VectorIterator< T >::operator-> () const [inline]
```

Apibrimas failo [Vector.h](#) eilutje 37.

5.31.4.13 operator<()

```
template<typename T>
bool VectorIterator< T >::operator< (
    const VectorIterator< T > & other) const [inline]
```

Apibrimas failo [Vector.h](#) eilutje 81.

5.31.4.14 operator<=()

```
template<typename T>
bool VectorIterator< T >::operator<= (
    const VectorIterator< T > & other) const [inline]
```

Apibrimas failo [Vector.h](#) eilutje 83.

5.31.4.15 operator==()

```
template<typename T>
bool VectorIterator< T >::operator== (
    const VectorIterator< T > & other) const    [inline]
```

Apibrimas failo [Vector.h](#) eilutje 79.

5.31.4.16 operator>()

```
template<typename T>
bool VectorIterator< T >::operator> (
    const VectorIterator< T > & other) const    [inline]
```

Apibrimas failo [Vector.h](#) eilutje 82.

5.31.4.17 operator>=()

```
template<typename T>
bool VectorIterator< T >::operator>= (
    const VectorIterator< T > & other) const    [inline]
```

Apibrimas failo [Vector.h](#) eilutje 84.

5.31.4.18 operator[]()

```
template<typename T>
reference VectorIterator< T >::operator[] (
    difference_type n) const    [inline]
```

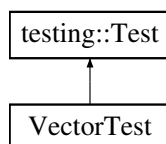
Apibrimas failo [Vector.h](#) eilutje 75.

Dokumentacija iai klasei sugeneruota i io failo:

- `header_files/Vector.h`

5.32 VectorTest Klas

Paveldimumo diagrama VectorTest:



Apsaugoti Atributai

- `Vector< int > v`

5.32.1 Smulkus apraymas

Apibrimas failo [test_students.cpp](#) eilutje 494.

5.32.2 Atribut Dokumentacija

5.32.2.1 v

```
Vector<int> VectorTest::v    [protected]
```

Apibrimas failo [test_students.cpp](#) eilutje 496.

Dokumentacija iai klasei sugeneruota i io failo:

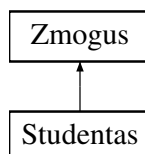
- `test_students.cpp`

5.33 Zmogus Klas

Abstrakti bazin klas bendram mogaus apraymui.

#include <Zmogus.h>

Paveldimumo diagrama Zmogus:



Viei Metodai

- `Zmogus (const string &v, const string &p)`
Parametrizuotas konstruktorius.
- `Zmogus ()`
Parametrizuotas konstruktorius su numatytosiomis reikmmis.
- `virtual Zmogus ()`
Virtualus destruktorius - reikalingas polimorfizmui.
- `string getVardas () const`
- `string getPavarde () const`
- `void setVardas (const string &v)`
Nustato mogaus vard
- `void setPavarde (const string &p)`
Nustato mogaus pavard
- `virtual string getInfo () const =0`
Grina mogaus informacij is metodus turi bti implementuotas visose ivestinese klasse.

Apsaugoti Metodai

- `virtual void paskaiciuoti ()=0`
Pagalbinis metodus mogaus duomenims paskaiiuoti Abstraktus metodus - turi bti implementuotas ives-
tinese klasse.

Apsaugoti Atributai

- `string vardas_`
- `string pavarde_`

Draugai

- `std::ostream & operator<< (std::ostream &os, const Zmogus &z)`
Ivesties operatorius - spausdina mogaus duomenis.

5.33.1 Smulkus apraymas

Abstrakti bazin klas bendram mogaus apraymui.

i klas nuo v1.5 versijos aprao bendr mogaus duomenis (vardas, pavard). Yra abstrakti - jos objekt kurti negalima, tik i jos ivestines klases.

Paveldi i ios klass: `Studentas`

Apibrimas failo `Zmogus.h` eilutje 17.

5.33.2 Konstruktoriaus ir Destruktoriaus Dokumentacija

5.33.2.1 Zmogus() [1/2]

Zmogus::Zmogus (
 const [string](#) & v,
 const [string](#) & p) [inline]
 Parametrizuotas konstruktorius.

Parametrai

v	mogaus vardas
p	mogaus pavard

Apibrimas failo [Zmogus.h](#) eilutje 34.
 Naudojamas [getInfo\(\)](#), ir [operator<<](#).

5.33.2.2 Zmogus() [2/2]

Zmogus::Zmogus () [inline]
 Parametrizuotas konstruktorius su numatytosiomis reikmmis.
 Apibrimas failo [Zmogus.h](#) eilutje 40.

5.33.2.3 Zmogus()

virtual Zmogus::Zmogus () [inline], [virtual]
 Virtualus destruktoriaus - reikalingas polimorfizmui.
 Apibrimas failo [Zmogus.h](#) eilutje 45.

5.33.3 Metod Dokumentacija

5.33.3.1 getInfo()

virtual [string](#) Zmogus::getInfo () const [pure virtual]
 Grina mogaus informacij is metodus turi bti implementuotas visose ivestinese klasse.
 Realizuota [Studentas](#).
 Ryiai [Zmogus\(\)](#).

5.33.3.2 getPavarde()

[string](#) Zmogus::getPavarde () const [inline]

Graina

mogaus pavard

Apibrimas failo [Zmogus.h](#) eilutje 56.
 Naudojamas [Studentas::getInfo\(\)](#).

5.33.3.3 getVardas()

[string](#) Zmogus::getVardas () const [inline]

Graina

mogaus vard

Apibrimas failo [Zmogus.h](#) eilutje 51.
 Naudojamas [Studentas::getInfo\(\)](#).

5.33.3.4 setPavarde()

```
void Zmogus::setPavarde (
    const string & p) [inline]
```

Nustato mogaus pavard

Apibrimas failo [Zmogus.h](#) eilutje 67.

5.33.3.5 setVardas()

```
void Zmogus::setVardas (
    const string & v) [inline]
```

Nustato mogaus vard

Apibrimas failo [Zmogus.h](#) eilutje 62.

5.33.4 Draugik Ir Susijusi Funkcij Dokumentacija

5.33.4.1 operator<<

```
std::ostream & operator<< (
    std::ostream & os,
    const Zmogus & z) [friend]
```

Ivesties operatorius - spausdina mogaus duomenis.

Parametrai

os	Ivesties srautas
z	mogaus objektas

Graina

Ivesties srautas

Apibrimas failo [Zmogus.cpp](#) eilutje 12.

Ryiai [operator<<](#), ir [Zmogus\(\)](#).

Naudojamas [operator<<](#).

5.33.5 Atribut Dokumentacija

5.33.5.1 pavarde_

```
string Zmogus::pavarde_ [protected]
```

Apibrimas failo [Zmogus.h](#) eilutje 20.

5.33.5.2 vardas_

```
string Zmogus::vardas_ [protected]
```

Apibrimas failo [Zmogus.h](#) eilutje 19.

Dokumentacija iai klasei sugeneruota i io failo:

- `header_files/Zmogus.h`

skyrius 6

Failo Dokumentacija

6.1 duomenų_valdymas.cpp

```
00001 #include <vector>
00002 #include <iostream>
00003 #include <string>
00004 #include <algorithm>
00005 #include <cctype>
00006 #include <limits>
00007 #include <random>
00008 #include <ctime>
00009 #include <fstream>
00010 #include <iomanip>
00011 #include <chrono>
00012 #include <stdexcept>
00013
00014 #include "../header_files/duomenų_valdymas.h"
00015 #include "../header_files/mat_funkcijos.h"
00016 #include "../header_files/output.h"
00017
00018 using std::cin;
00019 using std::cout;
00020 using std::string;
00021 using std::vector;
00022 using std::numeric_limits;
00023 using std::streamsize;
00024 using std::ifstream;
00025 using std::ofstream;
00026 using std::left;
00027 using std::setw;
00028 using std::fixed;
00029 using std::setprecision;
00030 using std::sort;
00031 using std::chrono::high_resolution_clock;
00032 using std::chrono::duration;
00033 using std::runtime_error;
00034 using std::exception;
00035
00036
00037
00038
00039 // Atsitiktinių skaičių generavimas
00040 void generuotiPaz(vector<Studentas>& grupe) {
00041     srand(time(0));
00042
00043     bool testiStudenta = true;
00044
00045     while (testiStudenta) {
00046         Studentas A;
00047
00048         bool vardasGeras = false;
00049         while (!vardasGeras) {
00050             cout << "Iveskite varda ir pavarde: ";
00051             cin >> std::ws; string v, p; cin >> v >> p; A.setVardas(v); A.setPavarde(p);
00052
00053             if (!arTikRaides(A.getVardas()) || !arTikRaides(A.getPavarde())) {
00054                 cout << "Klaida! Vardas ir pavarde turi buti sudaryti tik is raidziu!\n";
00055                 cin.clear();
00056             }
00057             else {
00058                 vardasGeras = true;
00059             }
00060         }
00061     }
```

```

00062     cout << string(80, '-') << "\n";
00063
00064     // Automatiskai generuojamas atsitiktinis pazymiu kiekis (3-10)
00065     int n = rand() % 8 + 3;
00066     A.reservePaz(n); // Atminties rezervacija pazymiams
00067
00068     // Automatiskai generuojami pazymiai
00069     cout << "Sugeneruota " << n << " pazymiu: ";
00070     for (int i = 0; i < n; i++) {
00071         int temp = generuotiPazymi();
00072         A.addPaz(temp);
00073         cout << temp << " ";
00074     }
00075     cout << "\n";
00076
00077     // Automatiskai generuojamas egzaminas
00078     A.setEgz(generuotiEgzamina());
00079     cout << "Egzamino ivertinimas: " << A.getEgz() << "\n";
00080     cout << string(80, '-') << "\n";
00081
00082     // Vidurkio skaiciavimas
00083     if (!A.isPazEmpty()) {
00084         double vid = vidurkis(A.getPaz());
00085         A.setRez(galutinisBalas(vid, A.getEgz()));
00086         // Medianos skaiciavimas
00087         double med = mediana(A.getPaz());
00088         A.setMed(galutinisBalas(med, A.getEgz()));
00089     }
00090     else {
00091         A.setRez(A.getEgz() * 0.6);
00092         A.setMed(A.getEgz() * 0.6);
00093     }
00094
00095     grupe.push_back(A);
00096
00097     // Klausimas ar testi
00098     char atsakymas;
00099     bool atsakymasTeisingas = false;
00100     while (!atsakymasTeisingas) {
00101         cout << "Ar norite generuoti dar viena studenta? (T/N): ";
00102         cin >> atsakymas;
00103         if (atsakymas == 'T' || atsakymas == 't') {
00104             testiStudenta = true;
00105             atsakymasTeisingas = true;
00106             cout << string(80, '-') << "\n";
00107         }
00108         else if (atsakymas == 'N' || atsakymas == 'n') {
00109             testiStudenta = false;
00110             atsakymasTeisingas = true;
00111         }
00112         else {
00113             cout << "Klaida! Iveskite T arba N!\n";
00114             cin.clear();
00115             cin.ignore(numeric_limits<streamsize>::max(), '\n');
00116         }
00117     }
00118 }
00119 }
00120
00121 void generuotiVardIrPav(vector<Studentas>& grupe) {
00122     srand(time(0));
00123
00124     try {
00125         // Nuskaitome failus
00126         vector<string> vyruVard, vyruPav, motVard, motPav;
00127         ifstream vvard("../vpv/vvard.txt");
00128         ifstream vpav("../vpv/vpav.txt");
00129         ifstream mvard("../vpv/mvard.txt");
00130         ifstream mpav("../vpv/mpav.txt");
00131
00132         // Patikrinimas ar failai atsidare
00133         if (!vvard.is_open() || !vpav.is_open() || !mvard.is_open() || !mpav.is_open()) {
00134             throw runtime_error("Nepavyko atidaryti vieno ar daugiau failu su vardais ir pavardemis!");
00135         }
00136
00137         // Nuskaitomi visi vardai ir pavardes
00138         string eilute;
00139         while (vvard >> eilute) vyruVard.push_back(eilute);
00140         while (vpav >> eilute) vyruPav.push_back(eilute);
00141         while (mvard >> eilute) motVard.push_back(eilute);
00142         while (mpav >> eilute) motPav.push_back(eilute);
00143
00144         vvard.close();
00145         vpav.close();
00146         mvard.close();
00147         mpav.close();
00148     }

```



```

00149     // Patikrinama ar failai ne tusti
00150     if (vyruVard.empty() || vyruPav.empty() || motVard.empty() || motPav.empty()) {
00151         throw runtime_error("Vienas ar daugiau failu yra tusti!");
00152     }
00153
00154     bool testiStudenta = true;
00155
00156     while (testiStudenta) {
00157         Studentas A;
00158
00159         // Atsitiktinai parenka lyti (0 - vyras, 1 - moteris)
00160         int lytis = rand() % 2;
00161
00162         // Generuojamas vardas ir pavarde pagal lyti
00163         if (lytis == 0) {
00164             int vardIndex = rand() % vyruVard.size();
00165             int pavIndex = rand() % vyruPav.size();
00166             A.setVardas(vyruVard[vardIndex]);
00167             A.setPavarde(vyruPav[pavIndex]);
00168         }
00169         else {
00170             int vardIndex = rand() % motVard.size();
00171             int pavIndex = rand() % motPav.size();
00172             A.setVardas(motVard[vardIndex]);
00173             A.setPavarde(motPav[pavIndex]);
00174         }
00175
00176         cout << "Sugeneruotas vardas ir pavarde: " << A.getVardas() << " " << A.getPavarde() << "\n";
00177         cout << string(80, '-') << "\n";
00178
00179         // Automatiškai generuojamas atsitiktinis pazymiu kiekis (3-10)
00180         int n = rand() % 8 + 3;
00181         A.reservePaz(n); // Atminties rezervacija pazymiams
00182
00183         // Automatiškai sugeneruojami pazymiai
00184         cout << "Sugeneruota " << n << " pazymiu: ";
00185         for (int i = 0; i < n; i++) {
00186             int temp = generuotiPazymi();
00187             A.addPaz(temp);
00188             cout << temp << " ";
00189         }
00190         cout << "\n";
00191
00192         // Automatiškai sugeneruojamas egzaminas
00193         A.setEgz(generuotiEgzamina());
00194         cout << "Egzamino ivertinimas: " << A.getEgz() << "\n";
00195         cout << string(80, '-') << "\n";
00196
00197         // Vidurkio skaiciavimas
00198         if (!A.isPazEmpty()) {
00199             double vid = vidurkis(A.getPaz());
00200             A.setRez(galutinisBalas(vid, A.getEgz()));
00201             // Medianos skaiciavimas
00202             double med = mediana(A.getPaz());
00203             A.setMed(galutinisBalas(med, A.getEgz()));
00204         }
00205         else {
00206             A.setRez(A.getEgz() * 0.6);
00207             A.setMed(A.getEgz() * 0.6);
00208         }
00209
00210         grupe.push_back(A);
00211
00212         // Klausimas ar testi
00213         char atsakymas;
00214         bool atsakymasTeisingas = false;
00215         while (!atsakymasTeisingas) {
00216             cout << "Ar norite generuoti dar viena studenta? (T/N): ";
00217             cin >> atsakymas;
00218             if (atsakymas == 'T' || atsakymas == 't') {
00219                 testiStudenta = true;
00220                 atsakymasTeisingas = true;
00221                 cout << string(80, '-') << "\n";
00222             }
00223             else if (atsakymas == 'N' || atsakymas == 'n') {
00224                 testiStudenta = false;
00225                 atsakymasTeisingas = true;
00226             }
00227             else {
00228                 cout << "Klaida! Iveskite T arba N!\n";
00229                 cin.clear();
00230                 cin.ignore(numeric_limits<streamsize>::max(), '\n');
00231             }
00232         }
00233     }
00234 }
00235 catch (const runtime_error& e) {

```

```

00236     cout << "Klaida generuojant studentus: " << e.what() << "\n";
00237 }
00238 catch (const exception& e) {
00239     cout << "Netiketa klaida: " << e.what() << "\n";
00240 }
00241 }
00242
00243 void skaitytiIsFailo(vector<Studentas>& grupe) {
00244     try {
00245         string failoPavadinimas;
00246         cout << "Iveskite failo pavadinima: ";
00247         cin >> failoPavadinimas;
00248
00249         // Priedamas testuojamas katalogas
00250         string kelias = "..\\studentai_test\\" + failoPavadinimas;
00251
00252         ifstream failas(kelias);
00253         if (!failas.is_open()) {
00254             throw runtime_error("Nepavyko atidaryti failo: " + kelias);
00255         }
00256
00257         // Praleidžiama antraste
00258         string eilute;
00259         getline(failas, eilute);
00260
00261         // Skaitomi studentai
00262         while (failas >> eilute) {
00263             Studentas A;
00264             A.setVardas(eilute);
00265             failas >> A.getPavarde();
00266
00267             vector<int> paz;
00268             paz.reserve(15); // Rezervuojama talpa (~15 pazymiu dzniausiai pakanka)
00269             int sk;
00270             while (failas >> sk) {
00271                 paz.push_back(sk);
00272                 if (failas.peek() == '\n') break;
00273             }
00274
00275             if (!paz.empty()) {
00276                 A.setEgz(paz.back());
00277                 paz.pop_back();
00278                 A.setPaz(paz);
00279                 A.setRez(galutinisBalas(vidurkis(A.getPaz()), A.getEgz()));
00280                 A.setMed(galutinisBalas(mediana(A.getPaz()), A.getEgz()));
00281                 grupe.push_back(A);
00282             }
00283             failas.ignore(numeric_limits<streamsize>::max(), '\n');
00284         }
00285
00286         failas.close();
00287
00288         if (grupe.empty()) {
00289             throw runtime_error("Failas yra tuscias arba neteisingai formatuotas!");
00290         }
00291
00292         cout << "Nuskaityta " << grupe.size() << " studentu!\n";
00293
00294         // Klausima kur isvesti rezultatus
00295         if (!grupe.empty()) {
00296             cout << string(80, '-') << "\n";
00297             cout << "Pasirinkite isvesties buda:\n";
00298             cout << "1. Ivesti i terminala\n";
00299             cout << "2. Irasyti i faila\n";
00300             int isvestiesPasirinkimas;
00301             while (!(cin >> isvestiesPasirinkimas) || (isvestiesPasirinkimas != 1 && isvestiesPasirinkimas != 2)) {
00302                 cout << "Klaida! Pasirinkite 1 arba 2: ";
00303                 cin.clear();
00304                 cin.ignore(numeric_limits<streamsize>::max(), '\n');
00305             }
00306             if (isvestiesPasirinkimas == 1) {
00307                 outputas(grupe);
00308             }
00309             else {
00310                 rasytIFaila(grupe);
00311             }
00312         }
00313     }
00314     catch (const runtime_error& e) {
00315         cout << "Klaida skaitant is failo: " << e.what() << "\n";
00316     }
00317     catch (const exception& e) {
00318         cout << "Netiketa klaida: " << e.what() << "\n";
00319     }
00320 }
00321
00322 void rasytIFaila(vector<Studentas>& grupe) {

```

```

00323     if (grupe.empty()) {
00324         cout << "Nera studentu duomenų!\n";
00325         return;
00326     }
00327
00328     // Klausima kaip rusiuoti
00329     cout << string(80, '-') << "\n";
00330     cout << "Pasirinkite rusiavimo buda:\n";
00331     cout << "1. Pagal vardą (A-Z)\n";
00332     cout << "2. Pagal pavardę (A-Z)\n";
00333     cout << "3. Pagal galutini bala (vidurki) - didejimo tvarka\n";
00334     cout << "4. Pagal galutini bala (mediana) - didejimo tvarka\n";
00335
00336     int pasirinkimas;
00337     while (!(cin > pasirinkimas) || pasirinkimas < 1 || pasirinkimas > 4) {
00338         cout << "Klaida! Iveskite skaiciu nuo 1 iki 4: ";
00339         cin.clear();
00340         cin.ignore(numeric_limits<streamsize>::max(), '\n');
00341     }
00342
00343     cout << "Pasirinktas rusiavimas: " << pasirinkimas << "\n";
00344
00345     // Pradedamas rusiavimo laiko matavimas
00346     auto pradzia = high_resolution_clock::now();
00347
00348     // Rusiuojama pagal pasirinkima
00349     switch (pasirinkimas) {
00350     case 1:
00351         sort(grupe.begin(), grupė.end(), [](const Studentas& a, const Studentas& b) {
00352             return a.getVardas() < b.getVardas();
00353         });
00354         break;
00355     case 2:
00356         sort(grupe.begin(), grupė.end(), [](const Studentas& a, const Studentas& b) {
00357             return a.getPavarde() < b.getPavarde();
00358         });
00359         break;
00360     case 3:
00361         sort(grupe.begin(), grupė.end(), [](const Studentas& a, const Studentas& b) {
00362             return a.getRez() < b.getRez();
00363         });
00364         break;
00365     case 4:
00366         sort(grupe.begin(), grupė.end(), [](const Studentas& a, const Studentas& b) {
00367             return a.getMed() < b.getMed();
00368         });
00369         break;
00370     default:
00371         cout << "Klaida! Neteisingas pasirinkimas. Rodoma be rusiavimo.\n";
00372         break;
00373     }
00374
00375     // Baigiamas rusiavimo laiko matavimas
00376     auto pabaiga = high_resolution_clock::now();
00377     duration<double> trukme = pabaiga - pradzia;
00378
00379     // Klausima failo pavadinimo
00380     string isvestiesFailas;
00381     cout << string(80, '-') << "\n";
00382     cout << "Iveskite isvesties failo pavadinima: ";
00383     cin > isvestiesFailas;
00384
00385     // Sukuriamas kelias i testuojama kataloga
00386     string kelias = "..\\studentai_test\\" + isvestiesFailas;
00387
00388     ofstream failas(kelias);
00389     if (!failas.is_open()) {
00390         cout << "Klaida! Nepavyko sukurti failo: " << kelias << "\n";
00391         return;
00392     }
00393
00394     // Antrastes eilute
00395     failas << left << setw(20) << "Vardas" << setw(20) << "Pavarde"
00396         << setw(20) << "Galutinis (Vid.)" << setw(20) << "Galutinis (Med.)\n";
00397     failas << string(80, '-') << "\n";
00398
00399     // Studentu duomenys
00400     for (const auto& A : grupė) {
00401         failas << left << setw(20) << A.getVardas() << setw(20) << A.getPavarde()
00402             << setw(20) << fixed << setprecision(2) << A.getRez()
00403             << setw(20) << fixed << setprecision(2) << A.getMed() << "\n";
00404     }
00405
00406     failas.close();
00407
00408     cout << string(80, '-') << "\n";
00409     cout << "Duomenys sekmingai irasyti i faila: " << kelias << "\n";

```

```
00410     cout << "Ivykdymo laikas: " << fixed << setprecision(7) << trukme.count() << " s\n";
00411 }
```

6.2 input.cpp

```
00001 #include <iostream>
00002 #include <limits>
00003
00004 #include "../header_files/input.h"
00005 #include "../header_files/mat_funkcijos.h"
00006
00007 using std::cin;
00008 using std::cout;
00009 using std::vector;
00010 using std::string;
00011 using std::numeric_limits;
00012 using std::streamsize;
00013
00014 void inputas(vector<Studentas>& grupe) {
00015     bool testiStudenta = true;
00016
00017     while (testiStudenta) {
00018         Studentas A;
00019
00020         bool vardasGeras = false;
00021         while (!vardasGeras) {
00022             cout << "Iveskite varda ir pavarde: ";
00023             cin >> std::ws; string v, p; cin >> v >> p; A.setVardas(v); A.setPavarde(p);
00024
00025             if (!arTikRaides(A.getVardas()) || !arTikRaides(A.getPavarde())) {
00026                 cout << "Klaida! Vardas ir pavarde turi buti sudaryti tik is raidziu!\n";
00027                 cin.clear();
00028             }
00029             else {
00030                 vardasGeras = true;
00031             }
00032         }
00033
00034         cout << string(80, '-') << "\n";
00035         cout << "Iveskite semestro ivertinimus (0-10). Iveskite -1 kad baigtumete: \n";
00036         A.reservePaz(15); // Rezervuojama vieta pazymiams (sumazina atminties reallokacijas)
00037         int temp;
00038         while (true) {
00039             cout << "Iveskite pazymi (arba -1 kad baigtumete): ";
00040             if (!(cin >> temp)) {
00041                 cout << "Klaida! Iveskite skaiciu!\n";
00042                 cin.clear();
00043                 cin.ignore(numeric_limits<streamsize>::max(), '\n');
00044             }
00045             else if (temp == -1) {
00046                 break;
00047             }
00048             else if (temp < 0 || temp > 10) {
00049                 cout << "Klaida! Pazymys turi buti nuo 0 iki 10!\n";
00050             }
00051             else {
00052                 A.addPaz(temp);
00053             }
00054         }
00055
00056         cout << string(80, '-') << "\n";
00057
00058         bool egzaminasTeisingas = false;
00059         while (!egzaminasTeisingas) {
00060             cout << "Iveskite egzamina (0-10): ";
00061             int e; if (!(cin >> e)) {
00062                 cout << "Klaida! Iveskite skaiciu!\n";
00063                 cin.clear();
00064                 cin.ignore(numeric_limits<streamsize>::max(), '\n');
00065             }
00066             else if (e < 0 || e > 10) {
00067                 cout << "Klaida! Egzaminas turi buti nuo 0 iki 10!\n";
00068             }
00069             else {
00070                 A.setEgz(e); egzaminasTeisingas = true;
00071             }
00072         }
00073         cout << string(80, '-') << "\n";
00074
00075         // Vidurkio skaiciavimas
00076         if (!A.isPazEmpty()) {
00077             double vid = vidurkis(A.getPaz());
00078             A.setRez(galutinisBalas(vid, A.getEgz()));
00079             // Medianos skaiciavimas
```

```

00080         double med = mediana(A.getPaz());
00081         A.setMed(galutinisBalas(med, A.getEgz()));
00082     }
00083     else {
00084         A.setRez(A.getEgz() * 0.6);
00085         A.setMed(A.getEgz() * 0.6);
00086     }
00087
00088     grupe.push_back(A);
00089     A.clearPaz();
00090
00091     // Klausimas ar testi rankini generavima
00092     char atsakymas;
00093     bool atsakymasTeisingas = false;
00094     while (!atsakymasTeisingas) {
00095         cout << "Ar norite ivesti dar viena studenta? (T/N): ";
00096         cin >> atsakymas;
00097         if (atsakymas == 'T' || atsakymas == 't') {
00098             testiStudenta = true;
00099             atsakymasTeisingas = true;
00100             cout << string(80, '-') << "\n";
00101         }
00102         else if (atsakymas == 'N' || atsakymas == 'n') {
00103             testiStudenta = false;
00104             atsakymasTeisingas = true;
00105         }
00106         else {
00107             cout << "Klaida! Iveskite T arba N!\n";
00108             cin.clear();
00109             cin.ignore(numeric_limits<streamsize>::max(), '\n');
00110         }
00111     }
00112 }
00113 }
00114

```

6.3 mat_funkcijos.cpp

```

00001 #include "../header_files/mat_funkcijos.h"
00002 #include <algorithm>
00003 #include <cctype>
00004 #include <ctime>
00005
00006 using std::sort;
00007
00008 // Funkcija patikrinti ar vardas/pavarde turi tik raides, neleidzia jokiu simboliu
00009 bool arTikRaides(const string& str) {
00010     if (str.empty()) return false;
00011     for (char c : str) {
00012         if (!std::isalpha(c)) {
00013             return false;
00014         }
00015     }
00016     return true;
00017 }
00018
00019 // Medianos funkcija
00020 double mediana(vector<int> paz) {
00021     if (paz.empty()) return 0.0;
00022
00023     vector<int> temp = paz;
00024     sort(temp.begin(), temp.end());
00025     int n = temp.size();
00026     if (n % 2 == 0) {
00027         return (temp[n / 2 - 1] + temp[n / 2]) / 2.0;
00028     }
00029     else {
00030         return temp[n / 2];
00031     }
00032 }
00033
00034 // Vidurkio skaiciavimas is pazymių vektoriaus
00035 double vidurkis(const vector<int>& paz) {
00036     if (paz.empty()) return 0.0;
00037     int sum = 0;
00038     for (int p : paz) {
00039         sum += p;
00040     }
00041     return sum * 1.0 / paz.size();
00042 }
00043
00044 // Galutinio balo skaiciavimas pagal vidurki/mediana ir egzamina
00045 double galutinisBalas(double vidMed, int egz) {
00046     return vidMed * 0.4 + egz * 0.6;

```

```

00047 }
00048
00049 // Generuoti atsitiktini pazymi nuo 0 iki 10
00050 int generuotiPazymi() {
00051     return rand() % 11;
00052 }
00053
00054 // Generuoti atsitiktini egzamina nuo 0 iki 10
00055 int generuotiEgzamina() {
00056     return rand() % 11;
00057 }
00058
00059 // Vector versions
00060 double mediana(Vector<int> paz) {
00061     if (paz.empty()) return 0.0;
00062
00063     Vector<int> temp = paz;
00064     sort(temp.begin(), temp.end());
00065     int n = temp.size();
00066     if (n % 2 == 0) {
00067         return (temp[n / 2 - 1] + temp[n / 2]) / 2.0;
00068     }
00069     else {
00070         return temp[n / 2];
00071     }
00072 }
00073
00074 // Vidurkio skaiciavimas is Vector pazymiu
00075 double vidurkis(const Vector<int>& paz) {
00076     if (paz.empty()) return 0.0;
00077     int sum = 0;
00078     for (int p : paz) {
00079         sum += p;
00080     }
00081     return sum * 1.0 / paz.size();
00082 }

```

6.4 menu.cpp

```

00001 #include <vector>
00002 #include <iostream>
00003 #include <string>
00004 #include <stdexcept>
00005
00006 #include "../header_files/menu.h"
00007 #include "../header_files/studentas.h"
00008 #include "../header_files/input.h"
00009 #include "../header_files/output.h"
00010 #include "../header_files/duomeniu_valdymas.h"
00011 #include "../header_files/testavimas.h"
00012
00013 using std::cout;
00014 using std::cin;
00015 using std::string;
00016 using std::vector;
00017 using std::invalid_argument;
00018 using std::exception;
00019
00020 void menu() {
00021     vector<Studentas>grupe;
00022     int pas; // Pasirinkimas
00023     bool testi = true;
00024
00025     while (testi) {
00026         cout << string(80, '-') << "\n";
00027         cout << "Studentu Rezultatu skaiciavimo aplikacija\n";
00028         cout << string(80, '-') << "\n";
00029         cout << "1. Ivesti duomenis ranka \n";
00030         cout << "2. Generuoti tik pazymius \n";
00031         cout << "3. Generuoti studentu vardus, pavardes ir pazymius \n";
00032         cout << "4. Nuskaityti duomenis is failo \n";
00033         cout << "5. Sukurti testavimo failus (1000 - 10000000 irasu)\n";
00034         cout << "6. Atlikti spartos analize (nuskaitymas, rusiavimas, dalijimas, isvedimas)\n";
00035         cout << "7. Baigti darba \n";
00036
00037         try {
00038             if (!(cin > pas)) {
00039                 throw invalid_argument("Neteisingas ivedimas! Iveskite skaiciu.");
00040             }
00041         }
00042         catch (const invalid_argument& e) {
00043             cout << "Klaida: " << e.what() << "\n";
00044             cin.clear();
00045             cin.ignore(1000, '\n');

```

```

00046         continue;
00047     }
00048
00049     // Pasirinkimo isvestis
00050     switch (pas) {
00051     case 1:
00052         try {
00053             cout << "1. Ivesti duomenisss ranka \n";
00054             inputas(grupe);
00055             outputas(grupe);
00056             grupe.clear();
00057         }
00058         catch (const exception& e) {
00059             cout << "Klaida vykdant 1 funkcija: " << e.what() << "\n";
00060             grupe.clear();
00061         }
00062         break;
00063     case 2:
00064         try {
00065             cout << "2. Generuoti tik pazymius \n";
00066             generuotiPaz(grupe);
00067             outputas(grupe);
00068             grupe.clear();
00069         }
00070         catch (const exception& e) {
00071             cout << "Klaida vykdant 2 funkcija: " << e.what() << "\n";
00072             grupe.clear();
00073         }
00074         break;
00075     case 3:
00076         try {
00077             cout << "3. Generuoti studentu vardus, pavardes ir pazymius \n";
00078             generuotiVardIrPav(grupe);
00079             outputas(grupe);
00080             grupe.clear();
00081         }
00082         catch (const exception& e) {
00083             cout << "Klaida vykdant 3 funkcija: " << e.what() << "\n";
00084             grupe.clear();
00085         }
00086         break;
00087     case 4:
00088         try {
00089             cout << "4. Nuskaityti duomenis is failo \n";
00090             skaitytiIsFailo(grupe);
00091             grupe.clear();
00092         }
00093         catch (const exception& e) {
00094             cout << "Klaida vykdant 4 funkcija: " << e.what() << "\n";
00095             grupe.clear();
00096         }
00097         break;
00098     case 5:
00099         try {
00100             cout << "5. Testavimo failu kurimas \n";
00101             sukurtiTestavimoFailus();
00102         }
00103         catch (const exception& e) {
00104             cout << "Klaida vykdant generavima: " << e.what() << "\n";
00105         }
00106         break;
00107     case 6:
00108         try {
00109             cout << "6. Spartos analizes vykdymas \n";
00110             atliktiSpartosAnalyze();
00111         }
00112         catch (const exception& e) {
00113             cout << "Klaida vykdant spartos analize: " << e.what() << "\n";
00114         }
00115         break;
00116     case 7:
00117         cout << "Programa uzdaroma \n";
00118         testi = false;
00119         break;
00120
00121     // Ivestis ivedus netinkama pasirinkima
00122     default:
00123         cout << "Klaida! Pasirinkite skaiciu nuo 1 iki 7 \n";
00124         cin.clear();
00125         cin.ignore(1000, '\n');
00126         break;
00127     }
00128 }
00129 }

```

6.5 output.cpp

```

00001 #include <iostream>
00002 #include <iomanip>
00003 #include <algorithm>
00004 #include <limits>
00005 #include <chrono>
00006
00007 #include "../header_files/output.h"
00008
00009 using std::cin;
00010 using std::cout;
00011 using std::string;
00012 using std::vector;
00013 using std::left;
00014 using std::right;
00015 using std::setw;
00016 using std::setprecision;
00017 using std::sort;
00018 using std::fixed;
00019 using std::numeric_limits;
00020 using std::streamsize;
00021 using std::chrono::high_resolution_clock;
00022 using std::chrono::duration;
00023
00024 void outputas(vector<Studentas>& grupe) {
00025     if (grupe.empty()) {
00026         cout << "Nera studentu duomenu!\n";
00027         return;
00028     }
00029
00030     // Klausiamė kaip rusiuoti
00031     cout << string(80, '-') << "\n";
00032     cout << "Pasirinkite rusiavimo buda:\n";
00033     cout << "1. Pagal vardą (A-Z)\n";
00034     cout << "2. Pagal pavardę (A-Z)\n";
00035     cout << "3. Pagal galutini bala (vidurki) - didejimo tvarka\n";
00036     cout << "4. Pagal galutini bala (mediana) - didejimo tvarka\n";
00037
00038     int pasirinkimas;
00039     while (!(cin > pasirinkimas) || pasirinkimas < 1 || pasirinkimas > 4) {
00040         cout << "Klaida! Iveskite skaiciu nuo 1 iki 4: ";
00041         cin.clear();
00042         cin.ignore(numeric_limits<streamsize>::max(), '\n');
00043     }
00044
00045     cout << "Pasirinktas rusiavimas: " << pasirinkimas << "\n";
00046
00047     // Pradedamas rusiavimo laiko matavimas
00048     auto pradzia = high_resolution_clock::now();
00049
00050     // Rusiuojama pagal pasirinkima
00051     switch (pasirinkimas) {
00052     case 1:
00053         sort(grupe.begin(), grupe.end(), comparePagalVarda);
00054         break;
00055     case 2:
00056         sort(grupe.begin(), grupe.end(), comparePagalPavarde);
00057         break;
00058     case 3:
00059         sort(grupe.begin(), grupe.end(), comparePagalReza);
00060         break;
00061     case 4:
00062         sort(grupe.begin(), grupe.end(), comparePagalMeda);
00063         break;
00064     default:
00065         cout << "Klaida! Neteisingas pasirinkimas. Rodoma be rusiavimo.\n";
00066         break;
00067     }
00068
00069     // Baigiamas rusiavimo laiko matavimas
00070     auto pabaiga = high_resolution_clock::now();
00071     duration<double> trukme = pabaiga - pradzia;
00072
00073     // Antrastes eilute
00074     cout << string(80, '-') << "\n";
00075     cout << left << setw(20) << "Vardas" << setw(20) << "Pavarde"
00076         << setw(20) << "Galutinis (Vid.)" << setw(20) << "Galutinis (Med.)\n";
00077
00078     // Studentu duomenys
00079     for (const auto& A : grupe) {
00080         cout << left << setw(20) << A.getVardas() << setw(20) << A.getPavarde()
00081             << setw(20) << fixed << setprecision(2) << A.getRez()
00082             << setw(20) << fixed << setprecision(2) << A.getMed() << "\n";
00083     }
00084

```



```

00085     cout << string(80, '-') << "\n";
00086     cout << "Ivykdyimo laikas: " << fixed << setprecision(7) << trukme.count() << " s\n";
00087 }

```

6.6 Studentas.cpp

```

00001 #include "../header_files/Studentas.h"
00002 #include "../header_files/mat_funkcijos.h"
00003 #include <iostream>
00004
00005 using std::istream;
00006
00007 // Nuskaitymas is srauto konstruktoriuje
00008 Studentas::Studentas(std::istream& is) : Zmogus(), egz_(0), rez_(0.0), med_(0.0) {
00009     readStudent(is);
00010 }
00011
00012 // ===== RULE OF FIVE =====
00013
00014 // Destruktorius
00015 Studentas::~Studentas() {
00016     paz_.clear();
00017 }
00018
00019 // Copy konstruktorius
00020 Studentas::Studentas(const Studentas& other)
00021 : Zmogus(other.vardas_, other.pavarde_, paz_(other.paz_),
00022     egz_(other.egz_), rez_(other.rez_), med_(other.med_)) {
00023 }
00024
00025 // Copy assignment operator
00026 Studentas& Studentas::operator=(const Studentas& other) {
00027     if (this != &other) {
00028         setVardas(other.getVardas());
00029         setPavarde(other.getPavarde());
00030         paz_ = other.paz_;
00031         egz_ = other.egz_;
00032         rez_ = other.rez_;
00033         med_ = other.med_;
00034     }
00035     return *this;
00036 }
00037
00038 // Move konstruktorius
00039 Studentas::Studentas(Studentas&& other) noexcept
00040 : Zmogus(std::move(other.vardas_), std::move(other.pavarde_),
00041     paz_(std::move(other.paz_)), egz_(other.egz_), rez_(other.rez_),
00042     med_(other.med_)) {
00043     other.setEgz(0);
00044     other.setRez(0.0);
00045     other.setMed(0.0);
00046     other.paz_.clear();
00047 }
00048
00049 // Move assignment operator
00050 Studentas& Studentas::operator=(Studentas&& other) noexcept {
00051     if (this != &other) {
00052         paz_.clear();
00053         setVardas(std::move(other.vardas_));
00054         setPavarde(std::move(other.pavarde_));
00055         paz_ = std::move(other.paz_);
00056         egz_ = other.egz_;
00057         rez_ = other.rez_;
00058         med_ = other.med_;
00059         other.setEgz(0);
00060         other.setRez(0.0);
00061         other.setMed(0.0);
00062         other.paz_.clear();
00063     }
00064     return *this;
00065 }
00066
00067 // Implementacija abstraktaus metodo i Zmogus
00068 string Studentas::getInfo() const {
00069     return getVardas() + " " + getPavarde();
00070 }
00071
00072 // Privati pagalbine funkcija
00073 void Studentas::paskaiciuotiGalutinius() {
00074     if (!paz_.empty()) {
00075         double vid = vidurkis(paz_);
00076         double med = mediana(paz_);
00077         rez_ = galutinisBalas(vid, egz_);
00078         med_ = galutinisBalas(med, egz_);

```

```

00079     } else {
00080         rez_ = egz_ * 0.6;
00081         med_ = egz_ * 0.6;
00082     }
00083 }
00084
00085 // Implementacija abstraktaus metodo paskaiciuoti() i Zmogus
00086 void Studentas::paskaiciuoti() {
00087     paskaiciuotiGalutinius();
00088 }
00089
00090 // Duomenu nuskaitymas ir apskaiciavimas
00091 std::istream& Studentas::readStudent(std::istream& is) {
00092     string vardas, pavarde;
00093     is >> vardas >> pavarde;
00094     if (!is) return is;
00095     setVardas(vardas);
00096     setPavarde(pavarde);
00097     paz_.clear();
00098     int p;
00099     for (int i = 0; i < 5; i++) {
00100         is >> p;
00101         paz_.push_back(p);
00102     }
00103     is >> egz_;
00104     paskaiciuotiGalutinius();
00105     return is;
00106 }
00107
00108 // ===== I/O OPERATORIAI =====
00109
00110 // Ivesties operatorius
00111 std::ostream& operator<<(std::ostream& os, const Studentas& s) {
00112     os << "Vardas: " << s.getVardas() << " | Pavarde: " << s.getPavarde() << " | ";
00113     os << "Egzaminas: " << s.egz_ << " | Vidurkis rezultatas: " << s.rez_ << " | ";
00114     os << "Mediana rezultatas: " << s.med_ << " | Pazymiai: ";
00115
00116     if (!s.paz_.empty()) {
00117         for (int paz : s.paz_) {
00118             os << paz << " ";
00119         }
00120     } else {
00121         os << "(nera)";
00122     }
00123
00124     return os;
00125 }
00126
00127 // ?vesties operatorius - nuskaity student? i srauto
00128 std::istream& operator>>(std::istream& is, Studentas& s) {
00129     return s.readStudent(is);
00130 }
00131
00132 // ===== LYGINIMO FUNKCIJOS =====
00133
00134 bool comparePagalVarda(const Studentas& a, const Studentas& b) {
00135     return a.getVardas() < b.getVardas();
00136 }
00137
00138 bool comparePagalPavarde(const Studentas& a, const Studentas& b) {
00139     return a.getPavarde() < b.getPavarde();
00140 }
00141
00142 bool comparePagalReza(const Studentas& a, const Studentas& b) {
00143     return a.getRez() < b.getRez();
00144 }
00145
00146 bool comparePagalMeda(const Studentas& a, const Studentas& b) {
00147     return a.getMed() < b.getMed();
00148 }

```

6.7 testavimas.cpp

```

00001 #include <iostream>
00002 #include <fstream>
00003 #include <vector>
00004 #include <string>
00005 #include <chrono>
00006 #include <iomanip>
00007 #include <random>
00008 #include <algorithm>
00009 #include <stdexcept>
00010 #include <list>
00011 #include <deque>

```

```

00012 #include <type_traits>
00013 #include <iterator> // std::make_move_iterator
00014
00015 #include "../header_files/testavimas.h"
00016 #include "../header_files/mat_funkcijos.h"
00017 #include "../header_files/Vector.h"
00018
00019 using std::cout;
00020 using std::cin;
00021 using std::vector;
00022 using std::string;
00023 using std::ofstream;
00024 using std::ifstream;
00025 using std::chrono::high_resolution_clock;
00026 using std::chrono::duration;
00027 using std::fixed;
00028 using std::setprecision;
00029 using std::left;
00030 using std::setw;
00031 using std::to_string;
00032 using std::move;
00033 using std::list;
00034 using std::deque;
00035
00036 const vector<int> SZ = { 1000, 10000, 100000, 1000000, 10000000 };
00037 const int paz_kiekis = 5;
00038
00039 void sukurtiTestavimoFailus() {
00040     int pas;
00041     cout << "\nPasirinkite, kuri faila norite sukurti:\n";
00042     cout << "1. 1000 irasu\n";
00043     cout << "2. 10000 irasu\n";
00044     cout << "3. 100000 irasu\n";
00045     cout << "4. 1 000000 irasu\n";
00046     cout << "5. 10000000 irasu\n";
00047     cout << "6. Visus auksciau isvardintus\n";
00048     cout << "Jusu pasirinkimas: ";
00049     cin >> pas;
00050
00051     vector<int> pasirinkti_SZ;
00052     if (pas == 1) pasirinkti_SZ = { 1000 };
00053     else if (pas == 2) pasirinkti_SZ = { 10000 };
00054     else if (pas == 3) pasirinkti_SZ = { 100000 };
00055     else if (pas == 4) pasirinkti_SZ = { 1000000 };
00056     else if (pas == 5) pasirinkti_SZ = { 10000000 };
00057     else if (pas == 6) pasirinkti_SZ = SZ;
00058     else {
00059         cout << "Neteisingas pasirinkimas!\n";
00060         return;
00061     }
00062
00063     srand(static_cast<unsigned>(time(nullptr)));
00064     cout << "Failu kurimas prasidejo...\n";
00065     cout << string(80, '-') << "\n";
00066
00067     for (int n : pasirinkti_SZ) {
00068         string fname = "Studentai_test\\studentai_" + to_string(n) + ".txt";
00069
00070         // Pradedamas matuoti laikas tiems irasams kurti
00071         auto start = high_resolution_clock::now();
00072
00073         ofstream out(fname);
00074
00075         // Antraste
00076         out << left << setw(20) << "Vardas" << setw(20) << "Pavarde";
00077         for (int i = 1; i <= paz_kiekis; i++) {
00078             out << setw(8) << ("ND" + to_string(i));
00079         }
00080         out << "Egz\n";
00081
00082         // Tiesiai irasome i faila (be vektoriui pagal uzduoties salyga)
00083         for (int i = 1; i <= n; i++) {
00084             out << left << setw(20) << ("Vardas" + to_string(i)) << setw(20) << ("Pavarde" + to_string(i));
00085             for (int k = 0; k < paz_kiekis; k++) {
00086                 out << setw(8) << (rand() % 10 + 1);
00087             }
00088             out << (rand() % 10 + 1) << "\n";
00089         }
00090         out.close();
00091
00092         // Stabdyti laikmati ir isvesti i ekrana laika
00093         auto end = high_resolution_clock::now();
00094         duration<double> diff = end - start;
00095
00096         cout << n << " irasu failo sukurimo laikas: " << fixed << setprecision(5) << diff.count() << " s.\n";
00097     }
00098 }

```

```

00099
00100 template <typename Container>
00101 void nuskaitytiDuomenis(const string& fname, Container& grupe) {
00102     ifstream in(fname);
00103     string header;
00104     std::getline(in, header);
00105
00106     Studentas st;
00107     st.reservePaz(paz_kiekis);
00108
00109     while (true) {
00110         string v, p;
00111         if (!(in >> v >> p)) break;
00112         st.setVardas(v);
00113         st.setPavarde(p);
00114         st.clearPaz();
00115         int pr;
00116         for (int i = 0; i < paz_kiekis; i++) {
00117             in >> pr;
00118             st.addPaz(pr);
00119         }
00120         int egz; in >> egz; st.setEgz(egz);
00121         st.setRez(galutinisBalas(vidurkis(st.getPaz()), st.getEgz()));
00122         grupe.push_back(st);
00123     }
00124     in.close();
00125 }
00126
00127 template <typename Container>
00128 double tirtiKontaineri(int n, const string& fname, int strat) {
00129     auto total_start = high_resolution_clock::now();
00130
00131     // 1. Duomenu nuskaitymas is failo
00132     auto start = high_resolution_clock::now();
00133
00134     Container grupe;
00135     if constexpr (std::is_same_v<Container, vector<Studentas> >) {
00136         try {
00137             grupe.reserve(n); // list neturi reserve() funkcijos
00138         } catch (const std::exception& e) {
00139             cout << "Nepakanka atminties rezervuoti " << n << " elementu!\n";
00140             return 0.0;
00141         }
00142     }
00143
00144     nuskaitytiDuomenis(fname, grupe);
00145
00146     auto end = high_resolution_clock::now();
00147     duration<double> ms_read = end - start;
00148     cout << n << " Irasu failo nuskaitymo laikas: " << fixed << setprecision(5) << ms_read.count() << " s.\n";
00149
00150     // 2. Rusiavimas
00151     start = high_resolution_clock::now();
00152     if constexpr (std::is_same_v<Container, list<Studentas> >) {
00153         grupe.sort([(const Studentas& a, const Studentas& b) {
00154             return a.getRez() < b.getRez();
00155         }]);
00156     } else {
00157         std::sort(grupe.begin(), grupe.end(), [(const Studentas& a, const Studentas& b) {
00158             return a.getRez() < b.getRez();
00159         }]);
00160     }
00161     end = high_resolution_clock::now();
00162     duration<double> ms_sort = end - start;
00163     cout << n << " Irasu rusiavimas didejimo tvarka: " << fixed << setprecision(5) << ms_sort.count() << " s.\n";
00164
00165     // 3. Skirstymas i dvi grupes
00166     start = high_resolution_clock::now();
00167     Container vargsiukai;
00168     Container kietiakai;
00169
00170     if (strat == 1) {
00171         if constexpr (std::is_same_v<Container, vector<Studentas> >) {
00172             vargsiukai.reserve(n / 2 + 100);
00173             kietiakai.reserve(n / 2 + 100);
00174         }
00175
00176         for (auto& s : grupe) {
00177             if (s.getRez() < 5.0) {
00178                 vargsiukai.push_back(move(s));
00179             }
00180             else {
00181                 kietiakai.push_back(move(s));
00182             }
00183         }
00184         Container().swap(grupe); // Isvalome pradini kontaineri
00185     }

```

```

00186     else if (strat == 2) {
00187         if constexpr (std::is_same_v<Container, vector<Studentas>) {
00188             // Sukrupuojame vektorių vietoje (O(N) laikas):
00189             auto splitPoint = std::partition(grupe.begin(), grupe.end(), [](const Studentas& s) {
00190                 return s.getRez() >= 5.0; // Salyga kietiškams
00191             });
00192
00193             // IŠ anksto rezervuojame atminties vargiukams (nebutina, bet dar labiau pagreitina)
00194             vargiukai.reserve(std::distance(splitPoint, grupe.end()));
00195
00196             // NAUDOJAME MOVE: Perkeliame vargiukus iš pagrindinio vektoriaus galo į naują.
00197             vargiukai.insert(vargiukai.end(),
00198                 std::make_move_iterator(splitPoint),
00199                 std::make_move_iterator(grupe.end()));
00200
00201             // Ištriname perkeltus elementus iš originalaus vektoriaus.
00202             grupe.erase(splitPoint, grupe.end());
00203             kietiškai = move(grupe); // Like grupeje yra kietiškai
00204         }
00205         else {
00206             auto it = grupe.begin();
00207             while (it != grupe.end()) {
00208                 if (it->getRez() < 5.0) {
00209                     vargiukai.push_back(move(*it));
00210                     it = grupe.erase(it); // vector/deque atveju tai leta, list atveju - greita
00211                 }
00212                 else {
00213                     ++it;
00214                 }
00215             }
00216             kietiškai = move(grupe); // Like grupeje yra kietiškai
00217         }
00218     }
00219     else if (strat == 3) {
00220         if constexpr (std::is_same_v<Container, list<Studentas>) {
00221             // list atveju efektyviausia yra iškirpti elementus per splice + remove if ar pagal sąlygą
00222             auto it = std::partition(grupe.begin(), grupe.end(), [](const Studentas& s) {
00223                 return s.getRez() < 5.0; // Vargiukaikeliauja į patį priekį
00224             });
00225             // Iškerpame vargiukus iš grupės tiesiai į vargiukų list be memory re-allocation
00226             vargiukai.splice(vargiukai.begin(), grupe, grupe.begin(), it);
00227             kietiškai = move(grupe);
00228         }
00229         else {
00230             // std::vector ir std::deque atveju efektyviausia naudoti std::partition in-place
00231             if constexpr (std::is_same_v<Container, vector<Studentas>) {
00232                 vargiukai.reserve(n / 2 + 100);
00233             }
00234             auto it = std::stable_partition(grupe.begin(), grupe.end(), [](const Studentas& s) {
00235                 return s.getRez() < 5.0;
00236             });
00237
00238             // Iteruojam per pradžia kur atsidurs vargiukai po stable_partition pakeitimą
00239             vargiukai.insert(vargiukai.end(), std::make_move_iterator(grupe.begin()), std::make_move_iterator(it));
00240             // Kas liko originalioje grupėje iteruojame į kietišką (nuo it iki end)
00241             kietiškai.insert(kietiškai.end(), std::make_move_iterator(it), std::make_move_iterator(grupe.end()));
00242
00243             Container().swap(grupe); // Sunaikinam originalą
00244         }
00245     }
00246
00247     end = high_resolution_clock::now();
00248     duration<double> ms_split = end - start;
00249     cout << n << " Irasu dalijimo laikas (" << strat << "-a strategija): " << fixed << setprecision(5) << ms_split.count() << " s.\n";
00250
00251     // 4.1 Irasome vargiukus į failą
00252     start = high_resolution_clock::now();
00253     ofstream outV("vargiukai\\vargiukai_" + to_string(n) + ".txt");
00254     outV << left << setw(20) << "Vardas" << setw(20) << "Pavarde" << "Galutinis (Vid.)\n";
00255     for (const auto& s : vargiukai) {
00256         outV << left << setw(20) << s.getVardas() << setw(20) << s.getPavarde() << fixed << setprecision(2) << s.getRez() << "\n";
00257     }
00258     outV.close();
00259     Container().swap(vargiukai); // Islaisviname RAM perrasydami!
00260     end = high_resolution_clock::now();
00261     duration<double> ms_write_v = end - start;
00262     cout << n << " Irasu 'vargiuku' irasymo laikas: " << fixed << setprecision(5) << ms_write_v.count() << " s.\n";
00263
00264     // 4.2 Irasome kietišką į failą
00265     start = high_resolution_clock::now();
00266     ofstream outK("kietiškai\\kietiškai_" + to_string(n) + ".txt");
00267     outK << left << setw(20) << "Vardas" << setw(20) << "Pavarde" << "Galutinis (Vid.)\n";
00268     for (const auto& s : kietiškai) {
00269         outK << left << setw(20) << s.getVardas() << setw(20) << s.getPavarde() << fixed << setprecision(2) << s.getRez() << "\n";
00270     }
00271     outK.close();
00272     Container().swap(kietiškai); // Islaisviname ir kietišką!

```

```

00273 end = high_resolution_clock::now();
00274 duration<double> ms_write_k = end - start;
00275 cout << n << " Irasu 'kietiaku' irasymo laikas: " << fixed << setprecision(5) << ms_write_k.count() << " s.\n";
00276
00277 // Bendras laikas
00278 auto total_end = high_resolution_clock::now();
00279 duration<double> ms_total = total_end - total_start;
00280 return ms_total.count();
00281 }
00282
00283 void atliktiSpartosAnalyze() {
00284     int pas, kont, strat, test_kartai;
00285     cout << "\nPasirinkite, su kuriuo failu norite atlikti spartos analize:\n";
00286     cout << "1. 1000 irasu failas\n";
00287     cout << "2. 10000 irasu failas\n";
00288     cout << "3. 100000 irasu failas\n";
00289     cout << "4. 1000000 irasu failas\n";
00290     cout << "5. 10000000 irasu failas\n";
00291     cout << "6. Visi failai is eiles\n";
00292     cout << "7. VECTOR TESTAVIMAS (100k, 1M)\n";
00293     cout << "Jusu pasirinkimas: ";
00294     cin >> pas;
00295
00296     if (pas == 7) {
00297         // Vector testing
00298         int strat_v, test_kartai_v;
00299         cout << "\nPasirinkite testavimo strategija:\n";
00300         cout << "1. 1 strategija (Sukurti 2 naujus konteinerius)\n";
00301         cout << "2. 2 strategija (Sukurti 1 nauja konteineri, trinti is pagrindinio)\n";
00302         cout << "Jusu pasirinkimas: ";
00303         cin >> strat_v;
00304
00305         cout << "\nIveskite, kiek kartu norite pakartoti testavima: ";
00306         cin >> test_kartai_v;
00307
00308         cout << "\nPradedama Vector spartos analize...\n";
00309
00310         vector<int> pasirinkti_SZ = { 100000, 1000000 };
00311
00312         for (int n : pasirinkti_SZ) {
00313             string fname = "Studentai_test\\studentai_" + to_string(n) + ".txt";
00314
00315             ifstream patikrinimas(fname);
00316             if (!patikrinimas.good()) {
00317                 cout << "Failo " << fname << " nera. Praleidžiama.\n";
00318                 continue;
00319             }
00320             patikrinimas.close();
00321
00322             cout << string(80, '-') << "\n";
00323             cout << n << " irasu Vector spartos analize\n";
00324             cout << string(80, '-') << "\n";
00325
00326             double visas_laikas = 0.0;
00327
00328             for (int i = 0; i < test_kartai_v; i++) {
00329                 cout << "nTESTO NUMERIS: " << i + 1 << "\n";
00330                 visas_laikas += tirtiKonteineri<Vector<Studentas>(n, fname, strat_v);
00331             }
00332
00333             cout << "\n=====n";
00334             cout << "Vidutinis Vector testo laikas po " << test_kartai_v << " bandymu:\n";
00335             cout << fixed << setprecision(5) << visas_laikas / test_kartai_v << " s.\n";
00336             cout << "=====n\n";
00337         }
00338         return;
00339     }
00340
00341     vector<int> pasirinkti_SZ;
00342     if (pas == 1) pasirinkti_SZ = { 1000 };
00343     else if (pas == 2) pasirinkti_SZ = { 10000 };
00344     else if (pas == 3) pasirinkti_SZ = { 100000 };
00345     else if (pas == 4) pasirinkti_SZ = { 1000000 };
00346     else if (pas == 5) pasirinkti_SZ = { 10000000 };
00347     else if (pas == 6) pasirinkti_SZ = SZ;
00348     else {
00349         cout << "Neteisingas pasirinkimas!\n";
00350         return;
00351     }
00352
00353     cout << "\nPasirinkite konteinerio tipa testavimui:\n";
00354     cout << "1. std::vector\n";
00355     cout << "2. std::list\n";
00356     cout << "3. std::deque\n";
00357     cout << "Jusu pasirinkimas: ";
00358     cin >> kont;
00359

```

```

00360 cout << "\nPasirinkite testavimo strategija:\n";
00361 cout << "1. 1 strategija (Sukurti 2 naujus konteinerius)\n";
00362 cout << "2. 2 strategija (Sukurti 1 nauja konteineri, trinti is pagrindinio)\n";
00363 cout << "3. 3 strategija (Optimizuoti algoritmu metodai)\n";
00364 cout << "Jusu pasirinkimas: ";
00365 cin >> strat;
00366
00367 cout << "\nIveskite, kiek kartu norite pakartoti testavima (laiku vidurkiui): ";
00368 cin >> test_kartai;
00369
00370 cout << "\nPradedama spartos analize pagal nurodytus zingsnius...\n";
00371
00372 for (int n : pasirinkti_SZ) {
00373     string fname = "Studentai_test\\studentai_" + to_string(n) + ".txt";
00374
00375     // Tikrinimas ar failas atsidaro nenaudojant filesystem
00376     ifstream patikrinimas(fname);
00377     if (!patikrinimas.good()) {
00378         cout << "Failo " << fname << " nera. Praleidžiama.\n";
00379         continue;
00380     }
00381     patikrinimas.close();
00382
00383     cout << string(80, '-') << "\n";
00384     cout << n << " Irasu spartos analize\n";
00385     cout << string(80, '-') << "\n";
00386
00387     double visas_laikas = 0.0;
00388
00389     for (int i = 0; i < test_kartai; i++) {
00390         cout << "\nTESTO NUMERIS: " << i + 1 << "\n";
00391         if (kont == 1) visas_laikas += tirtiKonteineri<vector<Studentas>(n, fname, strat);
00392         else if (kont == 2) visas_laikas += tirtiKonteineri<list<Studentas>(n, fname, strat);
00393         else if (kont == 3) visas_laikas += tirtiKonteineri<deque<Studentas>(n, fname, strat);
00394         else {
00395             cout << "Neteisingas konteinerio tipas!\n";
00396             return;
00397         }
00398     }
00399     cout << "\n===== \n";
00400     cout << "Vidutinis viso testo (nuskaitymas+dalijimas+isvedimas) laikas po " << test_kartai << " bandymu:\n";
00401     cout << fixed << setprecision(5) << visas_laikas / test_kartai << " s.\n";
00402     cout << "===== \n\n";
00403 }
00404 }
00405
00406 void PadalintiStudentusZabioGreiciu(std::vector<Studentas>& studentai, std::vector<Studentas>& vargsiukai) {
00407
00408     // 1. Sukrupuojame vektoriui vietoje (O(N) laikas):
00409     // Visi, kuriu balas >= 5.0 atsiduria priekyje, o < 5.0 - gale.
00410     // Jei jumps BUTINA iðlaikyti jau esama eiliðkuma (pvz., abecelini),
00411     // vietoj 'std::partition' naudokite 'std::stable_partition'.
00412     auto splitPoint = std::partition(studentai.begin(), studentai.end(), [(const Studentas& s) {
00413         return s.getRez() >= 5.0; // Pakeista is galutinis i rez
00414     }]);
00415
00416     // 2. Ið anksto rezervuojame atminties vargðiuokams (nebutina, bet dar labiau pagreitina)
00417     vargsiukai.reserve(std::distance(splitPoint, studentai.end()));
00418
00419     // 3. NAUDOJAME MOVE: Perkeliame vargðiuokus ið pagrindinio vektoriaus galo i naujaji.
00420     // std::make_move_iterator uðtikrina, kad tekstai ir kiti duomenys nebutu kopijuojami ið naujo.
00421     vargsiukai.insert(vargsiukai.end(),
00422         std::make_move_iterator(splitPoint),
00423         std::make_move_iterator(studentai.end()));
00424
00425     // 4. Iðtriname perkeltus elementus ið originalaus vektoriaus.
00426     // Kadangi triname IK GALO, jokio duomeni stumdymo nebelieka - operacija ivyksta per O(1).
00427     studentai.erase(splitPoint, studentai.end());
00428 }

```

6.8 Zmogus.cpp

```

00001 #include "../header_files/Zmogus.h"
00002 #include <iostream>
00003
00004 using std::ostream;
00005
00006 /**
00007  * @brief Ivesties operatorius - spausdina mogaus pagrindinius duomenis
00008  * @param os Ivesties srautas
00009  * @param z mogaus objektas
00010  * @return Ivesties srautas
00011  */
00012 std::ostream& operator<<(std::ostream& os, const Zmogus& z) {

```

```
00013     os « "Vardas: " « z.vardas_ « " | Pavarde: " « z.pavarde_;  
00014     return os;  
00015 }
```

6.9 duomenų_valdymas.h

```
00001 #pragma once  
00002  
00003 #include <vector>  
00004 #include <string>  
00005  
00006 #include "studentas.h"  
00007  
00008 using std::string;  
00009 using std::vector;  
00010  
00011 void generuotiPaz(vector<Studentas>& grupe);  
00012 void generuotiVardIrPav(vector<Studentas>& grupe);  
00013 void skaitytiIsFailo(vector<Studentas>& grupe);  
00014 void rasytiIFaila(vector<Studentas>& grupe);
```

6.10 input.h

```
00001 #pragma once  
00002  
00003 #include <vector>  
00004 #include <string>  
00005 #include "studentas.h"  
00006  
00007 using std::vector;  
00008 using std::string;  
00009  
00010 void inputas(vector<Studentas>& grupe);
```

6.11 Mat_funkcijos.h

```
00001 #pragma once  
00002  
00003 #include <string>  
00004 #include <vector>  
00005 #include "Vector.h"  
00006  
00007 using std::string;  
00008 using std::vector;  
00009  
00010 bool arTikRaides(const string& str);  
00011 double mediana(vector<int> paz);  
00012 double vidurkis(const vector<int>& paz); // Priima per referenc greiui  
00013 double galutinisBalas(double vidMed, int egz);  
00014 int generuotiPazymi();  
00015 int generuotiEgzamina();  
00016  
00017 // Vector versions  
00018 double mediana(Vector<int> paz);  
00019 double vidurkis(const Vector<int>& paz);  
00020
```

6.12 menu.h

```
00001 #pragma once  
00002  
00003 void menu();  
00004
```

6.13 output.h

```
00001 #pragma once  
00002  
00003 #include <vector>  
00004 #include <string>  
00005  
00006 #include "studentas.h"  
00007
```



```

00008 using std::vector;
00009 using std::string;
00010
00011 void outputas(vector<Studentas>& grupe);

```

6.14 Studentas.h

```

00001 #pragma once
00002
00003 #include <iostream>
00004 #include <string>
00005 #include "Vector.h"
00006 #include "Zmogus.h"
00007
00008 using std::string;
00009
00010 // Ivestin klas i Zmogus - aprao student
00011 // Paveldi mogaus duomenis (vardas, pavard) i Zmogus klass
00012 // Turi papildomus duomenis: paymius, egzamino rezultat ir galutinius rezultatus
00013 // Implementuoja Rule of Five
00014 class Studentas : public Zmogus {
00015 private:
00016     Vector<int> paz_;
00017     int egz_;
00018     double rez_;
00019     double med_;
00020
00021     // Privatas helperis skaiivimams
00022     void paskaiciuotiGalutinius();
00023
00024     // Implementacija abstraktaus metodo i Zmogus
00025     virtual void paskaiciuoti() override;
00026
00027 public:
00028     // Numatytasis konstruktorius
00029     Studentas() : Zmogus("A", "BB"), egz_(0), rez_(0.0), med_(0.0) {}
00030
00031     // Parametrizuotas konstruktorius
00032     Studentas(string v, string p, int e) : Zmogus(v, p), egz_(e), rez_(0.0), med_(0.0) {}
00033
00034     // Konstruktorius su nuskaitymu is srauto
00035     Studentas(std::istream& is);
00036
00037     // Rule of Five
00038
00039     // Destruktorius
00040     ~Studentas();
00041
00042     // Copy konstruktorius
00043     Studentas(const Studentas& other);
00044
00045     // Copy assignment operator
00046     Studentas& operator=(const Studentas& other);
00047
00048     // Move konstruktorius
00049     Studentas(Studentas&& other) noexcept;
00050
00051     // Move assignment operator
00052     Studentas& operator=(Studentas&& other) noexcept;
00053
00054     // Getters
00055     inline const Vector<int>& getPaz() const { return paz_; }
00056     inline Vector<int>& getPaz() { return paz_; }
00057     inline int getEgz() const { return egz_; }
00058     inline double getRez() const { return rez_; }
00059     inline double getMed() const { return med_; }
00060
00061     // Setters
00062     inline void setEgz(int e) { egz_ = e; }
00063     inline void setRez(double r) { rez_ = r; }
00064     inline void setMed(double m) { med_ = m; }
00065
00066     // Darbas su paymiai
00067     inline void addPaz(int p) { paz_.push_back(p); }
00068     inline void clearPaz() { paz_.clear(); }
00069     inline void reservePaz(size_t n) { paz_.reserve(n); }
00070     inline bool isPazEmpty() const { return paz_.empty(); }
00071     inline void setPaz(const Vector<int>& p) { paz_ = p; }
00072
00073     // Skaitymas ir skaiivimas
00074     std::istream& readStudent(std::istream& is);
00075
00076     // Implementacija abstraktaus metodo i Zmogus
00077     virtual string getInfo() const override;

```

```

00078
00079 // I/O operatoriai
00080 friend std::ostream& operator<<(std::ostream& os, const Studentas& s);
00081 friend std::istream& operator>>(std::istream& is, Studentas& s);
00082 };
00083
00084 // Ne klasės nars, bet su klase tiesiogiai susijusios lyginimo funkcijos
00085 bool comparePagalVarda(const Studentas& a, const Studentas& b);
00086 bool comparePagalPavarde(const Studentas& a, const Studentas& b);
00087 bool comparePagalReza(const Studentas& a, const Studentas& b);
00088 bool comparePagalMeda(const Studentas& a, const Studentas& b);
00089

```

6.15 testavimas.h

```

00001 #pragma once
00002 #include <vector>
00003 #include "studentas.h"
00004 #include "Vector.h"
00005
00006 void sukurtiTestavimoFailus();
00007 void atliktiSpartosAnalize();
00008 void atliktiSpartosAnalizeVectorVersion();

```

6.16 Vector.h

```

00001 #pragma once
00002
00003 #include <iostream>
00004 #include <stdexcept>
00005 #include <utility>
00006 #include <algorithm>
00007 #include <limits>
00008 #include <vector>
00009
00010 /**
00011 * @brief Random access iterator for Vector<T> container
00012 *
00013 * VectorIterator provides bidirectional and random access to Vector elements.
00014 * Supports all standard iterator operations including comparison, arithmetic,
00015 * and dereferencing.
00016 *
00017 * @tparam T The element type that the iterator points to
00018 *
00019 * @see Vector
00020 */
00021 template <typename T>
00022 class VectorIterator {
00023 public:
00024     using iterator_category = std::random_access_iterator_tag;
00025     using value_type = T;
00026     using difference_type = std::ptrdiff_t;
00027     using pointer = T*;
00028     using reference = T&;
00029
00030 private:
00031     T* _ptr;
00032
00033 public:
00034     VectorIterator(T* ptr = nullptr) : _ptr(ptr) {}
00035
00036     reference operator*() const { return *_ptr; }
00037     pointer operator->() const { return _ptr; }
00038
00039     VectorIterator& operator++() {
00040         ++_ptr;
00041         return *this;
00042     }
00043     VectorIterator operator++(int) {
00044         VectorIterator temp = *this;
00045         ++_ptr;
00046         return temp;
00047     }
00048
00049     VectorIterator& operator--() {
00050         --_ptr;
00051         return *this;
00052     }
00053     VectorIterator operator--(int) {
00054         VectorIterator temp = *this;
00055         --_ptr;

```

```

00056     return temp;
00057 }
00058
00059 VectorIterator operator+(difference_type n) const {
00060     return VectorIterator(_ptr + n);
00061 }
00062 VectorIterator operator-(difference_type n) const {
00063     return VectorIterator(_ptr - n);
00064 }
00065
00066 VectorIterator& operator+=(difference_type n) {
00067     _ptr += n;
00068     return *this;
00069 }
00070 VectorIterator& operator-=(difference_type n) {
00071     _ptr -= n;
00072     return *this;
00073 }
00074
00075 reference operator[](difference_type n) const {
00076     return _ptr[n];
00077 }
00078
00079 bool operator==(const VectorIterator& other) const { return _ptr == other._ptr; }
00080 bool operator!=(const VectorIterator& other) const { return _ptr != other._ptr; }
00081 bool operator<(const VectorIterator& other) const { return _ptr < other._ptr; }
00082 bool operator>(const VectorIterator& other) const { return _ptr > other._ptr; }
00083 bool operator<=(const VectorIterator& other) const { return _ptr <= other._ptr; }
00084 bool operator>=(const VectorIterator& other) const { return _ptr >= other._ptr; }
00085
00086 difference_type operator-(const VectorIterator& other) const {
00087     return _ptr - other._ptr;
00088 }
00089 };
00090
00091 /**
00092  * @class Vector
00093  * @brief Custom dynamic array container template
00094  *
00095  * Vector<T> is a generic container that stores elements of type T in a
00096  * dynamically allocated array. It provides O(1) amortized time complexity
00097  * for push_back(), while maintaining contiguous memory layout like std::vector.
00098  *
00099  * The class implements the Rule of Five pattern and fully supports deep copy
00100  * and move semantics.
00101  *
00102  * **Key Features:**
00103  * - Dynamic memory management (automatic resizing)
00104  * - Random access iterators (begin(), end(), rbegin(), rend())
00105  * - Copy and move semantics (Rule of Five)
00106  * - Standard container operations (push_back, pop_back, insert, erase, etc.)
00107  * - Comparable performance to std::vector for most operations
00108  *
00109  * **Memory Strategy:**
00110  * - When capacity is exhausted, reallocates with roughly 1.5x growth factor
00111  * - Tracks reallocation count for performance analysis
00112  * - Supports manual reserve() for optimization
00113  *
00114  * **Example Usage:**
00115  * @code
00116  * Vector<int> v;
00117  * v.push_back(10);
00118  * v.push_back(20);
00119  * v.reserve(100);    // Pre-allocate capacity
00120  *
00121  * for (int elem : v) {
00122  *     std::cout << elem << " ";
00123  * }
00124  * @endcode
00125  *
00126  * @tparam T Element type (must support copy and move semantics)
00127  *
00128  * @see VectorIterator
00129  * @see std::vector
00130  */
00131 template <typename T>
00132 class Vector {
00133 private:
00134     T* _data;
00135     size_t _size;
00136     size_t _capacity;
00137     size_t _realloc_count;
00138
00139     void reallocate(size_t new_capacity) {
00140         if (new_capacity < _size) {
00141             throw std::invalid_argument("New capacity must be >= current size");
00142         }
00143     }

```

```

00143     T* new_data = new T[new_capacity];
00144     for (size_t i = 0; i < _size; ++i) {
00145         new_data[i] = _data[i];
00146     }
00147     delete[] _data;
00148     _data = new_data;
00149     _capacity = new_capacity;
00150     ++_realloc_count;
00151 }
00152
00153 public:
00154 Vector() : _data(nullptr), _size(0), _capacity(0), _realloc_count(0) {}
00155 explicit Vector(size_t capacity) : _size(0), _capacity(capacity), _realloc_count(0) {
00156     _data = capacity > 0 ? new T[capacity] : nullptr;
00157 }
00158 Vector(size_t size, const T& value) : _size(size), _capacity(size), _realloc_count(0) {
00159     _data = size > 0 ? new T[size] : nullptr;
00160     for (size_t i = 0; i < size; ++i) _data[i] = value;
00161 }
00162
00163 // Conversion constructor from std::vector
00164 Vector(const std::vector<T>& other) : _size(other.size()), _capacity(other.size()), _realloc_count(0) {
00165     _data = _size > 0 ? new T[_size] : nullptr;
00166     for (size_t i = 0; i < _size; ++i) _data[i] = other[i];
00167 }
00168
00169 ~Vector() { delete[] _data; _data = nullptr; _size = 0; _capacity = 0; }
00170
00171 Vector(const Vector& other) : _size(other._size), _capacity(other._capacity), _realloc_count(0) {
00172     _data = _capacity > 0 ? new T[_capacity] : nullptr;
00173     for (size_t i = 0; i < _size; ++i) _data[i] = other._data[i];
00174 }
00175 Vector& operator=(const Vector& other) {
00176     if (this != &other) {
00177         delete[] _data;
00178         _size = other._size;
00179         _capacity = other._capacity;
00180         _realloc_count = 0;
00181         _data = _capacity > 0 ? new T[_capacity] : nullptr;
00182         for (size_t i = 0; i < _size; ++i) _data[i] = other._data[i];
00183     }
00184     return *this;
00185 }
00186
00187 Vector(Vector&& other) noexcept : _data(other._data), _size(other._size),
00188     _capacity(other._capacity), _realloc_count(other._realloc_count) {
00189     other._data = nullptr;
00190     other._size = 0;
00191     other._capacity = 0;
00192     other._realloc_count = 0;
00193 }
00194 Vector& operator=(Vector&& other) noexcept {
00195     if (this != &other) {
00196         delete[] _data;
00197         _data = other._data;
00198         _size = other._size;
00199         _capacity = other._capacity;
00200         _realloc_count = other._realloc_count;
00201         other._data = nullptr;
00202         other._size = 0;
00203         other._capacity = 0;
00204         other._realloc_count = 0;
00205     }
00206     return *this;
00207 }
00208
00209 inline size_t size() const { return _size; }
00210 inline size_t capacity() const { return _capacity; }
00211 inline bool empty() const { return _size == 0; }
00212
00213 inline T& operator[](size_t index) { return _data[index]; }
00214 inline const T& operator[](size_t index) const { return _data[index]; }
00215
00216 T& at(size_t index) {
00217     if (index >= _size) {
00218         throw std::out_of_range("Vector index out of range");
00219     }
00220     return _data[index];
00221 }
00222
00223 const T& at(size_t index) const {
00224     if (index >= _size) {
00225         throw std::out_of_range("Vector index out of range");
00226     }
00227     return _data[index];
00228 }
00229

```

```

00230 void clear() {
00231     _size = 0;
00232 }
00233
00234 T& front() {
00235     if (_size == 0) {
00236         throw std::out_of_range("Vector is empty: cannot access front()");
00237     }
00238     return _data[0];
00239 }
00240
00241 const T& front() const {
00242     if (_size == 0) {
00243         throw std::out_of_range("Vector is empty: cannot access front()");
00244     }
00245     return _data[0];
00246 }
00247
00248 T& back() {
00249     if (_size == 0) {
00250         throw std::out_of_range("Vector is empty: cannot access back()");
00251     }
00252     return _data[_size - 1];
00253 }
00254
00255 const T& back() const {
00256     if (_size == 0) {
00257         throw std::out_of_range("Vector is empty: cannot access back()");
00258     }
00259     return _data[_size - 1];
00260 }
00261
00262 void reserve(size_t new_capacity) {
00263     if (new_capacity > _capacity) {
00264         reallocate(new_capacity);
00265     }
00266 }
00267
00268 void shrink_to_fit() {
00269     if (_capacity > _size) {
00270         if (_size == 0) {
00271             delete[] _data;
00272             _data = nullptr;
00273             _capacity = 0;
00274         } else {
00275             reallocate(_size);
00276         }
00277     }
00278 }
00279
00280 void resize(size_t new_size) {
00281     if (new_size > _capacity) {
00282         reallocate(new_size);
00283     }
00284     _size = new_size;
00285 }
00286
00287 void resize(size_t new_size, const T& value) {
00288     if (new_size > _capacity) {
00289         reallocate(new_size);
00290     }
00291     if (new_size > _size) {
00292         for (size_t i = _size; i < new_size; ++i) {
00293             _data[i] = value;
00294         }
00295     }
00296     _size = new_size;
00297 }
00298
00299 void swap(Vector& other) noexcept {
00300     std::swap(_data, other._data);
00301     std::swap(_size, other._size);
00302     std::swap(_capacity, other._capacity);
00303     std::swap(_realloc_count, other._realloc_count);
00304 }
00305
00306 bool operator==(const Vector& other) const {
00307     if (_size != other._size) return false;
00308     for (size_t i = 0; i < _size; ++i) {
00309         if (_data[i] != other._data[i]) return false;
00310     }
00311     return true;
00312 }
00313
00314 bool operator!=(const Vector& other) const {
00315     return !(*this == other);
00316 }

```

```

00317
00318     bool operator<(const Vector& other) const {
00319         for (size_t i = 0; i < _size && i < other._size; ++i) {
00320             if (_data[i] < other._data[i]) return true;
00321             if (_data[i] > other._data[i]) return false;
00322         }
00323         return _size < other._size;
00324     }
00325
00326     bool operator>(const Vector& other) const {
00327         return other < *this;
00328     }
00329
00330     bool operator<=(const Vector& other) const {
00331         return !(other < *this);
00332     }
00333
00334     bool operator>=(const Vector& other) const {
00335         return !(*this < other);
00336     }
00337
00338     using iterator = VectorIterator<T>;
00339     using const_iterator = VectorIterator<const T>;
00340
00341     iterator begin() { return iterator(_data); }
00342     iterator end() { return iterator(_data + _size); }
00343
00344     const_iterator begin() const { return const_iterator(_data); }
00345     const_iterator end() const { return const_iterator(_data + _size); }
00346
00347     const_iterator cbegin() const { return const_iterator(_data); }
00348     const_iterator cend() const { return const_iterator(_data + _size); }
00349
00350     using reverse_iterator = std::reverse_iterator<iterator>;
00351     using const_reverse_iterator = std::reverse_iterator<const_iterator>;
00352
00353     reverse_iterator rbegin() { return reverse_iterator(end()); }
00354     reverse_iterator rend() { return reverse_iterator(begin()); }
00355
00356     const_reverse_iterator rbegin() const { return const_reverse_iterator(end()); }
00357     const_reverse_iterator rend() const { return const_reverse_iterator(begin()); }
00358
00359     const_reverse_iterator crbegin() const { return const_reverse_iterator(end()); }
00360     const_reverse_iterator crend() const { return const_reverse_iterator(begin()); }
00361
00362     void assign(size_t count, const T& value) {
00363         clear();
00364         if (count > _capacity) {
00365             reallocate(count);
00366         }
00367         for (size_t i = 0; i < count; ++i) {
00368             _data[i] = value;
00369         }
00370         _size = count;
00371     }
00372
00373     template<typename InputIt>
00374     void assign(InputIt first, InputIt last) {
00375         // This is a specialized version - only works with Vector iterators
00376         // For general case, user should use clear() + for loop
00377     }
00378
00379     iterator insert(const_iterator pos, const T& value) {
00380         size_t index = std::distance(cbegin(), pos);
00381         if (index > _size) {
00382             throw std::out_of_range("Iterator out of range");
00383         }
00384
00385         if (_size >= _capacity) {
00386             size_t new_capacity = (_capacity == 0) ? 1 : _capacity * 2;
00387             reallocate(new_capacity);
00388         }
00389
00390         for (size_t i = _size; i > index; --i) {
00391             _data[i] = _data[i - 1];
00392         }
00393         _data[index] = value;
00394         ++_size;
00395
00396         return iterator(_data + index);
00397     }
00398
00399     // Non-const overload for insert (accepts non-const iterator, treats as const_iterator)
00400     iterator insert(iterator pos, const T& value) {
00401         // pos is a mutable iterator, but we just use it as a position
00402         // Convert to size_t using cbegin()
00403

```

```

00404     size_t index = std::distance(begin(), pos);
00405     if (index > _size) {
00406         throw std::out_of_range("Iterator out of range");
00407     }
00408
00409     if (_size >= _capacity) {
00410         size_t new_capacity = (_capacity == 0) ? 1 : _capacity * 2;
00411         reallocate(new_capacity);
00412     }
00413
00414     for (size_t i = _size; i > index; --i) {
00415         _data[i] = _data[i - 1];
00416     }
00417     _data[index] = value;
00418     ++_size;
00419
00420     return iterator(_data + index);
00421 }
00422
00423 // Range insert overload with three arguments
00424 template<typename InputIt>
00425 iterator insert(const_iterator pos, InputIt first, InputIt last) {
00426     size_t index = std::distance(cbegin(), pos);
00427     if (index > _size) {
00428         throw std::out_of_range("Iterator out of range");
00429     }
00430
00431     // Count elements manually
00432     size_t count = 0;
00433     for (InputIt it = first; it != last; ++it) {
00434         ++count;
00435     }
00436
00437     if (_size + count > _capacity) {
00438         reallocate(std::max(_size + count, _capacity * 2));
00439     }
00440
00441     // Shift elements to the right
00442     for (size_t i = _size; i > index; --i) {
00443         _data[i + count - 1] = _data[i - 1];
00444     }
00445
00446     // Insert new elements
00447     size_t i = index;
00448     for (InputIt it = first; it != last; ++it, ++i) {
00449         _data[i] = *it;
00450     }
00451     _size += count;
00452
00453     return iterator(_data + index);
00454 }
00455
00456 // Range insert overload with non-const iterator
00457 template<typename InputIt>
00458 iterator insert(iterator pos, InputIt first, InputIt last) {
00459     size_t index = std::distance(begin(), pos);
00460     if (index > _size) {
00461         throw std::out_of_range("Iterator out of range");
00462     }
00463
00464     // Count elements manually
00465     size_t count = 0;
00466     for (InputIt it = first; it != last; ++it) {
00467         ++count;
00468     }
00469
00470     if (_size + count > _capacity) {
00471         reallocate(std::max(_size + count, _capacity * 2));
00472     }
00473
00474     // Shift elements to the right
00475     for (size_t i = _size; i > index; --i) {
00476         _data[i + count - 1] = _data[i - 1];
00477     }
00478
00479     // Insert new elements
00480     size_t i = index;
00481     for (InputIt it = first; it != last; ++it, ++i) {
00482         _data[i] = *it;
00483     }
00484     _size += count;
00485
00486     return iterator(_data + index);
00487 }
00488
00489 iterator erase(const_iterator pos) {
00490     size_t index = std::distance(cbegin(), pos);

```

```

00491     if (index >= _size) {
00492         throw std::out_of_range("Iterator out of range");
00493     }
00494
00495     for (size_t i = index; i < _size - 1; ++i) {
00496         _data[i] = _data[i + 1];
00497     }
00498     --_size;
00499
00500     return iterator(_data + index);
00501 }
00502
00503 iterator erase(const_iterator first, const_iterator last) {
00504     size_t first_index = std::distance(cbegin(), first);
00505     size_t last_index = std::distance(cbegin(), last);
00506
00507     if (first_index > _size || last_index > _size || first_index > last_index) {
00508         throw std::out_of_range("Invalid range");
00509     }
00510
00511     size_t count = last_index - first_index;
00512     for (size_t i = first_index; i < _size - count; ++i) {
00513         _data[i] = _data[i + count];
00514     }
00515     _size -= count;
00516
00517     return iterator(_data + first_index);
00518 }
00519
00520 // Non-const overload for erase (single element)
00521 iterator erase(iterator pos) {
00522     size_t index = std::distance(begin(), pos);
00523     if (index >= _size) {
00524         throw std::out_of_range("Iterator out of range");
00525     }
00526
00527     for (size_t i = index; i < _size - 1; ++i) {
00528         _data[i] = _data[i + 1];
00529     }
00530     --_size;
00531
00532     return iterator(_data + index);
00533 }
00534
00535 // Non-const overload for erase (range)
00536 iterator erase(iterator first, iterator last) {
00537     size_t first_index = std::distance(begin(), first);
00538     size_t last_index = std::distance(begin(), last);
00539
00540     if (first_index > _size || last_index > _size || first_index > last_index) {
00541         throw std::out_of_range("Invalid range");
00542     }
00543
00544     size_t count = last_index - first_index;
00545     for (size_t i = first_index; i < _size - count; ++i) {
00546         _data[i] = _data[i + count];
00547     }
00548     _size -= count;
00549
00550     return iterator(_data + first_index);
00551 }
00552
00553 template<typename... Args>
00554 void emplace_back(Args&&... args) {
00555     if (_size >= _capacity) {
00556         size_t new_capacity = (_capacity == 0) ? 1 : _capacity * 2;
00557         reallocate(new_capacity);
00558     }
00559     new (_data + _size) T(std::forward<Args>(args)...);
00560     ++_size;
00561 }
00562
00563 template<typename... Args>
00564 iterator emplace(const_iterator pos, Args&&... args) {
00565     size_t index = std::distance(cbegin(), pos);
00566     if (index > _size) {
00567         throw std::out_of_range("Iterator out of range");
00568     }
00569
00570     if (_size >= _capacity) {
00571         size_t new_capacity = (_capacity == 0) ? 1 : _capacity * 2;
00572         reallocate(new_capacity);
00573     }
00574
00575     for (size_t i = _size; i > index; --i) {
00576         _data[i] = _data[i - 1];
00577     }

```



```

00578     new (_data + index) T(std::forward<Args>(args)...);
00579     ++_size;
00580
00581     return iterator(_data + index);
00582 }
00583
00584 size_t max_size() const {
00585     return std::numeric_limits<size_t>::max() / sizeof(T);
00586 }
00587
00588 inline T* data() { return _data; }
00589 inline const T* data() const { return _data; }
00590
00591 inline size_t get_reallocation_count() const { return _realloc_count; }
00592 inline void reset_reallocation_count() { _realloc_count = 0; }
00593
00594 /**
00595  * @brief Add element to the end of the vector (copy semantics)
00596  *
00597  * Appends a copy of the given value to the end. If the current size equals
00598  * capacity, the vector doubles its capacity and reallocates.
00599  *
00600  * **Time Complexity:** O(1) amortized
00601  *
00602  * @param value The value to append (copied)
00603  *
00604  * @throws std::bad_alloc if memory allocation fails
00605  *
00606  * @see push_back(T&&), pop_back(), emplace_back()
00607  *
00608  * @example
00609  * @code
00610  * Vector<int> v;
00611  * v.push_back(42);
00612  * v.push_back(100);
00613  * @endcode
00614  */
00615 void push_back(const T& value) {
00616     if (_size >= _capacity) {
00617         size_t new_capacity = (_capacity == 0) ? 1 : _capacity * 2;
00618         reallocate(new_capacity);
00619     }
00620     _data[_size] = value;
00621     ++_size;
00622 }
00623
00624 /**
00625  * @brief Add element to the end of the vector (move semantics)
00626  *
00627  * Appends an rvalue reference to the vector. The element is moved
00628  * (not copied) into the vector. This is more efficient for expensive-to-copy types.
00629  *
00630  * **Time Complexity:** O(1) amortized
00631  *
00632  * @param value The rvalue reference to append (moved)
00633  *
00634  * @throws std::bad_alloc if memory allocation fails
00635  *
00636  * @see push_back(const T&), pop_back(), emplace_back()
00637  *
00638  * @example
00639  * @code
00640  * Vector<std::string> v;
00641  * v.push_back(std::string("Hello")); // Moves the temporary
00642  * @endcode
00643  */
00644 void push_back(T&& value) {
00645     if (_size >= _capacity) {
00646         size_t new_capacity = (_capacity == 0) ? 1 : _capacity * 2;
00647         reallocate(new_capacity);
00648     }
00649     _data[_size] = std::move(value);
00650     ++_size;
00651 }
00652
00653 /**
00654  * @brief Remove the last element from the vector
00655  *
00656  * Decrements the size by one, effectively removing the last element.
00657  * Does nothing if the vector is empty. Note: the element destructor
00658  * is NOT called (similar to std::vector::pop_back for POD types).
00659  *
00660  * **Time Complexity:** O(1) constant
00661  *
00662  * **Precondition:** Empty vector is safe to call on (no-op)
00663  *
00664  * @see push_back(), front(), back()

```

```

00665      *
00666      * @example
00667      * @code
00668      * Vector<int> v = {1, 2, 3};
00669      * v.pop_back(); // Now size is 2, last element is 2
00670      * @endcode
00671      */
00672      void pop_back() {
00673          if (_size > 0) {
00674              --_size;
00675          }
00676      }
00677 };
00678
00679
00680
00681

```

6.17 Zmogus.h

```

00001 #pragma once
00002
00003 #include <iostream>
00004 #include <string>
00005
00006 using std::string;
00007
00008 /**
00009  * @class Zmogus
00010  * @brief Abstrakti bazin klas bendram mogaus apraymui
00011  *
00012  * i klas nuo v1.5 versijos aprao bendr mogaus duomenis (vardas, pavard).
00013  * Yra abstrakti - jos objekt kurti negalima, tik i jos investines klases.
00014  *
00015  * Paveldi i ios klas: Studentas
00016  */
00017 class Zmogus {
00018 protected:
00019     string vardas_;
00020     string pavarde_;
00021
00022     /**
00023      * @brief Pagalbinis metodas mogaus duomenims paskaiiuoti
00024      * Abstraktus metodas - turi bti implementuotas investinse klase
00025      */
00026     virtual void paskaiciuoti() = 0;
00027
00028 public:
00029     /**
00030      * @brief Parametrizuotas konstruktorius
00031      * @param v mogaus vardas
00032      * @param p mogaus pavard
00033      */
00034     Zmogus(const string& v, const string& p)
00035         : vardas_(v), pavarde_(p) {}
00036
00037     /**
00038      * @brief Parametrizuotas konstruktorius su numatytosiomis reikmmis
00039      */
00040     Zmogus() : vardas_("Asmuo"), pavarde_("Asmenys") {}
00041
00042     /**
00043      * @brief Virtualus destruktoriaus - reikalingas polimorfizmui
00044      */
00045     virtual ~Zmogus() {}
00046
00047     // ===== GETTERS =====
00048     /**
00049      * @return mogaus vard
00050      */
00051     inline string getVardas() const { return vardas_; }
00052
00053     /**
00054      * @return mogaus pavard
00055      */
00056     inline string getPavarde() const { return pavarde_; }
00057
00058     // ===== SETTERS =====
00059     /**
00060      * @brief Nustato mogaus vard
00061      */
00062     inline void setVardas(const string& v) { vardas_ = v; }
00063
00064     /**

```

```

00065     * @brief Nustato mogaus pavarde
00066     */
00067 inline void setPavarde(const string& p) { pavarde_ = p; }
00068
00069 // ===== ABSTRAKTUS METODAS =====
00070 /**
00071  * @brief Grina mogaus informacij
00072  * is metodas turi bti implementuotas visose ivestins klasse
00073  */
00074 virtual string getInfo() const = 0;
00075
00076 // ===== I/O OPERATORIAI (friend functions) =====
00077 /**
00078  * @brief Ivesties operatorius - spausdina mogaus duomenis
00079  */
00080 friend std::ostream& operator<<(std::ostream& os, const Zmogus& z);
00081 };

```

6.18 objekt1uzd.cpp

```

00001 #include "header_files/menu.h"
00002
00003 int main() {
00004     menu();
00005     return 0;
00006 }

```

6.19 test_studentas.cpp

```

00001 #include <gtest/gtest.h>
00002 #include <iostream>
00003 #include <sstream>
00004 #include <cassert>
00005 #include <vector>
00006 #include "header_files/Studentas.h"
00007 #include "header_files/Vector.h"
00008
00009 using std::cout;
00010 using std::cin;
00011 using std::endl;
00012 using std::string;
00013 using std::vector;
00014 using std::stringstream;
00015
00016 int testai_praletii = 0;
00017 int testai_nepraletii = 0;
00018
00019 void printTestHeader(const string& testName) {
00020     cout << "\n" << string(60, '=') << endl;
00021     cout << "TEST: " << testName << endl;
00022     cout << string(60, '=') << endl;
00023 }
00024
00025 void testPassed(const string& info) {
00026     cout << "[PRIIMTAS] " << info << endl;
00027     testai_praletii++;
00028 }
00029
00030 void testFailed(const string& info) {
00031     cout << "[NEPRIIMTAS] " << info << endl;
00032     testai_nepraletii++;
00033 }
00034
00035 // TEST #1: Default konstruktorius
00036 void test_default_constructor() {
00037     printTestHeader("Default Konstruktorius");
00038
00039     Studentas s;
00040
00041     assert(s.getVardas() == "A");
00042     testPassed("Vardas inicijuotas i 'A'");
00043
00044     assert(s.getPavarde() == "BB");
00045     testPassed("Pavarde inicijuota i 'BB'");
00046
00047     assert(s.getEgz() == 0);
00048     testPassed("Egzaminas inicijuotas i 0");
00049
00050     assert(s.getRez() == 0.0);
00051     testPassed("Rezultatas inicijuotas i 0.0");
00052 }

```

```

00053     assert(s.getMed() == 0.0);
00054     testPassed("Mediana inicijuota i 0.0");
00055
00056     assert(s.isPazEmpty());
00057     testPassed("Pazymiai sarasas tucias");
00058 }
00059
00060 // TEST #2: Parametrizuotas konstruktorius
00061 void test_parametrized_constructor() {
00062     printTestHeader("Parametrizuotas Konstruktorius");
00063
00064     Studentas s("Jonas", "Jonaitis", 95);
00065
00066     assert(s.getVardas() == "Jonas");
00067     testPassed("Vardas teisingai priskirtas");
00068
00069     assert(s.getPavarde() == "Jonaitis");
00070     testPassed("Pavarde teisingai priskirta");
00071
00072     assert(s.getEgz() == 95);
00073     testPassed("Egzaminas teisingai priskirtas");
00074 }
00075
00076 // TEST #3: Copy konstruktorius
00077 void test_copy_constructor() {
00078     printTestHeader("Copy Konstruktorius");
00079
00080     Studentas s1("Petras", "Petrauskas", 85);
00081     s1.addPaz(8);
00082     s1.addPaz(9);
00083     s1.addPaz(10);
00084
00085     // Kuriam kopija
00086     Studentas s2 = s1;
00087
00088     assert(s2.getVardas() == s1.getVardas());
00089     testPassed("Kopijoje vardas sutampa");
00090
00091     assert(s2.getPavarde() == s1.getPavarde());
00092     testPassed("Kopijoje pavarde sutampa");
00093
00094     assert(s2.getEgz() == s1.getEgz());
00095     testPassed("Kopijoje egzaminas sutampa");
00096
00097     assert(s2.getPaz().size() == s1.getPaz().size());
00098     testPassed("Kopijoje pazymiu kiekis sutampa");
00099
00100     assert(s2.getPaz()[0] == s1.getPaz()[0]);
00101     testPassed("Kopijoje pazymiu vertes sutampa");
00102
00103     // Tikriname, kad tai skirtingi objektai (deep copy)
00104     Studentas s3 = s1;
00105     s3.addPaz(100);
00106     assert(s1.getPaz().size() != s3.getPaz().size());
00107     testPassed("Deep copy veikia - modifikacija nepaveikia originalo");
00108 }
00109
00110 // TEST #4: Copy assignment operator
00111 void test_copy_assignment() {
00112     printTestHeader("Copy Assignment Operatorius");
00113
00114     Studentas s1("Marta", "Martine", 92);
00115     s1.addPaz(7);
00116     s1.addPaz(8);
00117     s1.addPaz(9);
00118
00119     Studentas s2;
00120     s2 = s1;
00121
00122     assert(s2.getVardas() == "Marta");
00123     testPassed("Priskyrimo metu vardas is naujo priskirtas");
00124
00125     assert(s2.getPavarde() == "Martine");
00126     testPassed("Priskyrimo metu pavarde is naujo priskirta");
00127
00128     assert(s2.getPaz().size() == 3);
00129     testPassed("Priskyrimo metu pazymiai is naujo priskirti");
00130
00131     // Self-assignment check
00132     s2 = s2;
00133     assert(s2.getVardas() == "Marta");
00134     testPassed("Self-assignment saugiai veikia");
00135 }
00136
00137 // TEST #5: Move konstruktorius
00138 void test_move_constructor() {
00139     printTestHeader("Move Konstruktorius");

```

```

00140
00141     Studentas s1("Laima", "Laima", 88);
00142     s1.addPaz(6);
00143     s1.addPaz(7);
00144     s1.addPaz(8);
00145
00146     // Move konstruktorius
00147     Studentas s2 = std::move(s1);
00148
00149     assert(s2.getVardas() == "Laima");
00150     testPassed("Move konstruktorius - vardas is naujo priskirtas");
00151
00152     assert(s2.getPaz().size() == 3);
00153     testPassed("Move konstruktorius - pazymiai is naujo priskirti");
00154
00155     // s1 turetu tureti tuscius pazymius
00156     assert(s1.isPazEmpty());
00157     testPassed("Move konstruktorius - saltinis istustintas");
00158 }
00159
00160 // TEST #6: Move assignment operatorius
00161 void test_move_assignment() {
00162     printTestHeader("Move Assignment Operatorius");
00163
00164     Studentas s1("Ruta", "Rute", 90);
00165     s1.addPaz(5);
00166     s1.addPaz(6);
00167     s1.addPaz(7);
00168
00169     Studentas s2;
00170     s2 = std::move(s1);
00171
00172     assert(s2.getVardas() == "Ruta");
00173     testPassed("Move assignment - vardas is naujo priskirtas");
00174
00175     assert(s2.getPaz().size() == 3);
00176     testPassed("Move assignment - pazymiai is naujo priskirti");
00177
00178     assert(s1.isPazEmpty());
00179     testPassed("Move assignment - saltinis istustintas");
00180
00181     // Self-move assignment check
00182     s2 = std::move(s2);
00183     assert(s2.getVardas() == "Ruta");
00184     testPassed("Move assignment self-assignment saugiai veikia");
00185 }
00186
00187 // TEST #7: Destruktorius - RAII principas
00188 void test_destructor() {
00189     printTestHeader("Destruktorius - RAII Principas");
00190
00191     {
00192         Studentas s("Vidmantas", "Vidmante", 87);
00193         s.addPaz(9);
00194         s.addPaz(10);
00195         s.addPaz(8);
00196
00197         assert(s.getPaz().size() == 3);
00198         testPassed("Destruktoriaus testas - objektas sukurtas");
00199     } // Cia is automatiškai kviečiamas destruktoriaus
00200
00201     testPassed("Destruktor veikia be problemu (nera memory leaks)");
00202 }
00203
00204 // TEST #8: Isvesties operatorius (operator«)
00205 void test_output_operator() {
00206     printTestHeader("Isvesties Operatorius (operator«)");
00207
00208     Studentas s("Darius", "Darius", 93);
00209     s.addPaz(9);
00210     s.addPaz(10);
00211     s.addPaz(9);
00212
00213     stringstream ss;
00214     ss << s;
00215     string output = ss.str();
00216
00217     assert(output.find("Vardas: Darius") != string::npos);
00218     testPassed("Isvestis - vardas rastas");
00219
00220     assert(output.find("Pavarde: Darius") != string::npos);
00221     testPassed("Isvestis - pavarde rastas");
00222
00223     assert(output.find("Egzaminas: 93") != string::npos);
00224     testPassed("Isvestis - egzaminas rastas");
00225
00226     cout << "Isvesties pavyzdys: " << endl;

```

```

00227     cout << " " << s << endl;
00228 }
00229
00230 // TEST #9: Ivesties operatorius (operator»)
00231 void test_input_operator() {
00232     printTestHeader("Ivesties Operatorius (operator»)");
00233
00234     stringstream ss;
00235     ss << "Vaidas Vaidauskas 10 10 10 10 100";
00236
00237     Studentas s;
00238     ss >> s;
00239
00240     assert(s.getVardas() == "Vaidas");
00241     testPassed("Ivestis - vardas nuskaitytas");
00242
00243     assert(s.getPavarde() == "Vaidauskas");
00244     testPassed("Ivestis - pavarde nuskaityta");
00245
00246     assert(s.getEgz() == 100);
00247     testPassed("Ivestis - egzaminas nuskaitytas");
00248
00249     assert(s.getPaz().size() == 5);
00250     testPassed("Ivestis - pazymiai nuskaityti");
00251
00252     assert(s.getPaz()[0] == 10);
00253     testPassed("Ivestis - pazymio reiksme teisingai nuskaityta");
00254 }
00255
00256 // TEST #10: I/O operatoriai kartu (Round-trip test)
00257 void test_io_roundtrip() {
00258     printTestHeader("I/O Round-Trip Testas");
00259
00260     Studentas s1("Kristina", "Kristina", 96);
00261     s1.addPaz(8);
00262     s1.addPaz(9);
00263     s1.addPaz(10);
00264     s1.addPaz(9);
00265     s1.addPaz(8);
00266
00267     stringstream ss;
00268     ss << s1.getVardas() << " " << s1.getPavarde();
00269     for (int paz : s1.getPaz()) {
00270         ss << " " << paz;
00271     }
00272     ss << " " << s1.getEgz();
00273
00274     Studentas s2;
00275     ss >> s2;
00276
00277     assert(s2.getVardas() == s1.getVardas());
00278     testPassed("Round-trip - vardas saugai issaugotas ir nuskaitytas");
00279
00280     assert(s2.getPavarde() == s1.getPavarde());
00281     testPassed("Round-trip - pavarde saugai issaugotas ir nuskaitytas");
00282
00283     assert(s2.getPaz() == s1.getPaz());
00284     testPassed("Round-trip - pazymiai saugai issaugoti ir nuskaityti");
00285 }
00286
00287 // TEST #11: Modifikavimas po kopijavimo
00288 void test_modification_after_copy() {
00289     printTestHeader("Modifikavimas po kopijavimo");
00290
00291     Studentas s1("Elena", "Elene", 85);
00292     s1.addPaz(7);
00293     s1.addPaz(8);
00294
00295     Studentas s2 = s1;
00296
00297     s2.setEgz(95);
00298     assert(s1.getEgz() == 85 && s2.getEgz() == 95);
00299     testPassed("Egzamino modifikacija - originalas nepaveiktas");
00300
00301     s2.addPaz(9);
00302     assert(s1.getPaz().size() == 2 && s2.getPaz().size() == 3);
00303     testPassed("Pazymio pridejimas - originalas nepaveiktas");
00304 }
00305
00306 // TEST #12: Getter/Setter operaciju konsekvencija
00307 void test_getters_setters() {
00308     printTestHeader("Getters/Setters Konsekvencija");
00309
00310     Studentas s;
00311
00312     s.setVardas("Tomas");
00313     assert(s.getVardas() == "Tomas");

```

```

00314     testPassed("Vardas setter/getter veikia");
00315
00316     s.setPavarde("Tomas");
00317     assert(s.getPavarde() == "Tomas");
00318     testPassed("Pavarde setter/getter veikia");
00319
00320     s.setEgz(100);
00321     assert(s.getEgz() == 100);
00322     testPassed("Egzaminas setter/getter veikia");
00323
00324     s.setRez(90.5);
00325     assert(s.getRez() == 90.5);
00326     testPassed("Rezultatas setter/getter veikia");
00327
00328     s.setMed(88.0);
00329     assert(s.getMed() == 88.0);
00330     testPassed("Mediana setter/getter veikia");
00331 }
00332
00333 // TEST #13: Vector operacijos su pazymiais
00334 void test_paz_operations() {
00335     printTestHeader("Vector Operacijos su Pazymiais");
00336
00337     Studentas s;
00338
00339     assert(s.isPazEmpty());
00340     testPassed("isPazEmpty() grazina true tusciam sarakui");
00341
00342     s.addPaz(10);
00343     s.addPaz(9);
00344     assert(!s.isPazEmpty());
00345     testPassed("isPazEmpty() grazina false ne tusciam sarakui");
00346
00347     assert(s.getPaz().size() == 2);
00348     testPassed("Pazymiai pridedami teisingai");
00349
00350     s.clearPaz();
00351     assert(s.isPazEmpty());
00352     testPassed("clearPaz() isvalo pazymius");
00353 }
00354
00355 // ===== PAPILDOMI GOOGLE TEST TESTAI (v2.0) =====
00356
00357 // TEST #14: Vektoriaus su Studentais operacijos
00358 TEST(StudentasVectorTest, PushBackCopySemantics) {
00359     vector<Studentas> students;
00360
00361     Studentas s1("Audra", "Audre", 91);
00362     s1.addPaz(9);
00363     s1.addPaz(8);
00364
00365     students.push_back(s1); // Copy semantika
00366
00367     EXPECT_EQ(students.size(), 1);
00368     EXPECT_EQ(students[0].getVardas(), "Audra");
00369     EXPECT_EQ(students[0].getPaz().size(), 2);
00370 }
00371
00372 // TEST #15: Move semantika su vektoriumi
00373 TEST(StudentasVectorTest, PushBackMoveSemantics) {
00374     vector<Studentas> students;
00375
00376     Studentas s1("Giedra", "Giedre", 88);
00377     s1.addPaz(8);
00378
00379     students.push_back(std::move(s1)); // Move semantika
00380
00381     EXPECT_EQ(students.size(), 1);
00382     EXPECT_TRUE(s1.isPazEmpty()); // Originalas turi bti tuias
00383     EXPECT_EQ(students[0].getVardas(), "Giedra");
00384 }
00385
00386 // TEST #16: Deep copy - independenti duplikacija
00387 TEST(StudentasDeepCopyTest, IndependentCopies) {
00388     Studentas original("Laimas", "Laimas", 92);
00389     original.addPaz(9);
00390     original.addPaz(10);
00391
00392     Studentas copy1 = original;
00393     Studentas copy2 = original;
00394
00395     // Modifikuojame copy1
00396     copy1.setEgz(100);
00397     copy1.addPaz(8);
00398
00399     // Patikrinti, kad originalas ir copy2 nepakeisti
00400     EXPECT_EQ(original.getEgz(), 92);

```

```

00401 EXPECT_EQ(original.getPaz().size(), 2);
00402 EXPECT_EQ(copy2.getEgz(), 92);
00403 EXPECT_EQ(copy2.getPaz().size(), 2);
00404 EXPECT_EQ(copy1.getEgz(), 100);
00405 EXPECT_EQ(copy1.getPaz().size(), 3);
00406 }
00407
00408 // TEST #17: Self-assignment saugumas
00409 TEST(StudentasSelfAssignmentTest, CopySafetyCheck) {
00410     Studentas s("Monika", "Monika", 95);
00411     s.addPaz(10);
00412     s.addPaz(9);
00413
00414     // Self-assignment - turi bti saugus
00415     s = s;
00416
00417     EXPECT_EQ(s.getVardas(), "Monika");
00418     EXPECT_EQ(s.getEgz(), 95);
00419     EXPECT_EQ(s.getPaz().size(), 2);
00420 }
00421
00422 // TEST #18: Operatoriaus lankstumo - chaining
00423 TEST(StudentasOperatorTest, AssignmentChaining) {
00424     Studentas s1("Vincas", "Vincas", 92);
00425     Studentas s2("Petras", "Petrauskas", 88);
00426     Studentas s3;
00427
00428     // Assignment chaining: (s3 = s2) = s1;
00429     // Step 1: s3 = s2 grina s3 referenc, s3 becomes "Petras"
00430     // Step 2: (s3 referencija) = s1 priskyrimo s1 reikm s3
00431     (s3 = s2) = s1;
00432
00433     EXPECT_EQ(s3.getVardas(), "Vincas"); // s3 final value is Vincas
00434     EXPECT_EQ(s2.getVardas(), "Petras"); // s2 remains Petras
00435     EXPECT_EQ(s1.getVardas(), "Vincas"); // s1 remains Vincas
00436 }
00437
00438 // ===== ORIGINAL MAIN (su backward compatibility) =====
00439 int main(int argc, char** argv) {
00440     // Google Test inicializacija
00441     ::testing::InitGoogleTest(&argc, argv);
00442
00443     // Originals testai (v1.5)
00444     cout << "\n" << string(60, '*') << endl;
00445     cout << " STUDENTAS KLASIS VISOS METODU TESTAI (v1.5)" << endl;
00446     cout << " + GOOGLE TEST PAPILDYMAS (v2.0)" << endl;
00447     cout << string(60, '*') << endl;
00448
00449     try {
00450         test_default_constructor();
00451         test_parametrized_constructor();
00452         test_copy_constructor();
00453         test_copy_assignment();
00454         test_move_constructor();
00455         test_move_assignment();
00456         test_destructor();
00457         test_output_operator();
00458         test_input_operator();
00459         test_io_roundtrip();
00460         test_modification_after_copy();
00461         test_getters_setters();
00462         test_paz_operations();
00463
00464         cout << "\n" << string(60, '*') << endl;
00465         cout << "TESTO REZULTATAI:" << endl;
00466         cout << "Priimti testai: " << testai_praletii << endl;
00467         cout << "Nepriimti testai: " << testai_nepraletii << endl;
00468         cout << string(60, '*') << endl;
00469
00470         if (testai_nepraletii == 0) {
00471             cout << "\nVISI TESTAI PRIIMTI!" << endl;
00472         } else {
00473             cout << "\nKai kurie testai nepraejo!" << endl;
00474             return 1;
00475         }
00476     }
00477     catch (const std::exception& e) {
00478         cout << "Kritine klaida: " << e.what() << endl;
00479         return 1;
00480     }
00481
00482     cout << "\n" << string(60, '*') << endl;
00483     cout << "Vykdome GOOGLE TEST testus..." << endl;
00484     cout << string(60, '*') << "\n" << endl;
00485
00486     // Vykdyti Google Test testus
00487     return RUN_ALL_TESTS();

```



```

00488 }
00489
00490 // =====
00491 // VECTOR GTEST TESTAI
00492 // =====
00493
00494 class VectorTest : public ::testing::Test {
00495 protected:
00496     Vector<int> v;
00497 };
00498
00499 // Test 1: Default konstruktorius
00500 TEST_F(VectorTest, DefaultConstructor) {
00501     EXPECT_EQ(v.size(), 0);
00502     EXPECT_EQ(v.capacity(), 0);
00503     EXPECT_TRUE(v.empty());
00504 }
00505
00506 // Test 2: Parametrizuotas konstruktorius su size
00507 TEST_F(VectorTest, ConstructorWithCapacity) {
00508     Vector<int> v2(10);
00509     EXPECT_EQ(v2.size(), 0);
00510     EXPECT_EQ(v2.capacity(), 10);
00511 }
00512
00513 // Test 3: Parametrizuotas konstruktorius su size ir value
00514 TEST_F(VectorTest, ConstructorWithSizeAndValue) {
00515     Vector<int> v2(5, 42);
00516     EXPECT_EQ(v2.size(), 5);
00517     EXPECT_EQ(v2.capacity(), 5);
00518     EXPECT_EQ(v2[0], 42);
00519     EXPECT_EQ(v2[4], 42);
00520 }
00521
00522 // Test 4: push_back()
00523 TEST_F(VectorTest, PushBack) {
00524     v.push_back(10);
00525     v.push_back(20);
00526     v.push_back(30);
00527     EXPECT_EQ(v.size(), 3);
00528     EXPECT_EQ(v[0], 10);
00529     EXPECT_EQ(v[1], 20);
00530     EXPECT_EQ(v[2], 30);
00531 }
00532
00533 // Test 5: front() ir back()
00534 TEST_F(VectorTest, FrontAndBack) {
00535     v.push_back(10);
00536     v.push_back(20);
00537     v.push_back(30);
00538     EXPECT_EQ(v.front(), 10);
00539     EXPECT_EQ(v.back(), 30);
00540 }
00541
00542 // Test 6: pop_back()
00543 TEST_F(VectorTest, PopBack) {
00544     v.push_back(10);
00545     v.push_back(20);
00546     v.push_back(30);
00547     v.pop_back();
00548     EXPECT_EQ(v.size(), 2);
00549     EXPECT_EQ(v.back(), 20);
00550 }
00551
00552 // Test 7: at() su bounds checking
00553 TEST_F(VectorTest, AtWithBoundsChecking) {
00554     v.push_back(10);
00555     v.push_back(20);
00556     EXPECT_EQ(v.at(0), 10);
00557     EXPECT_EQ(v.at(1), 20);
00558     EXPECT_THROW(v.at(2), std::out_of_range);
00559 }
00560
00561 // Test 8: operator[]
00562 TEST_F(VectorTest, OperatorBrackets) {
00563     v.push_back(10);
00564     v.push_back(20);
00565     v[0] = 100;
00566     v[1] = 200;
00567     EXPECT_EQ(v[0], 100);
00568     EXPECT_EQ(v[1], 200);
00569 }
00570
00571 // Test 9: clear()
00572 TEST_F(VectorTest, Clear) {
00573     v.push_back(10);
00574     v.push_back(20);

```

```

00575     v.clear();
00576     EXPECT_EQ(v.size(), 0);
00577     EXPECT_TRUE(v.empty());
00578 }
00579
00580 // Test 10: reserve()
00581 TEST_F(VectorTest, Reserve) {
00582     v.reserve(100);
00583     EXPECT_GE(v.capacity(), 100);
00584     EXPECT_EQ(v.size(), 0);
00585 }
00586
00587 // Test 11: shrink_to_fit()
00588 TEST_F(VectorTest, ShrinkToFit) {
00589     v.push_back(10);
00590     v.push_back(20);
00591     v.push_back(30);
00592     v.shrink_to_fit();
00593     EXPECT_EQ(v.capacity(), v.size());
00594 }
00595
00596 // Test 12: resize()
00597 TEST_F(VectorTest, Resize) {
00598     v.push_back(10);
00599     v.push_back(20);
00600     v.resize(5);
00601     EXPECT_EQ(v.size(), 5);
00602
00603     v.resize(2);
00604     EXPECT_EQ(v.size(), 2);
00605 }
00606
00607 // Test 13: resize() su value
00608 TEST_F(VectorTest, ResizeWithValue) {
00609     v.resize(3, 42);
00610     EXPECT_EQ(v.size(), 3);
00611     EXPECT_EQ(v[0], 42);
00612     EXPECT_EQ(v[1], 42);
00613     EXPECT_EQ(v[2], 42);
00614 }
00615
00616 // Test 14: swap()
00617 TEST_F(VectorTest, Swap) {
00618     Vector<int> v2;
00619     v.push_back(10);
00620     v.push_back(20);
00621     v2.push_back(100);
00622
00623     v.swap(v2);
00624     EXPECT_EQ(v.size(), 1);
00625     EXPECT_EQ(v[0], 100);
00626     EXPECT_EQ(v2.size(), 2);
00627     EXPECT_EQ(v2[0], 10);
00628 }
00629
00630 // Test 15: operator== ir operator!=
00631 TEST_F(VectorTest, ComparisonOperators) {
00632     Vector<int> v2;
00633     v.push_back(10);
00634     v.push_back(20);
00635     v2.push_back(10);
00636     v2.push_back(20);
00637
00638     EXPECT_TRUE(v == v2);
00639     EXPECT_FALSE(v != v2);
00640
00641     v2.push_back(30);
00642     EXPECT_FALSE(v == v2);
00643     EXPECT_TRUE(v != v2);
00644 }
00645
00646 // Test 16: operator< ir operator>
00647 TEST_F(VectorTest, LessGreaterOperators) {
00648     Vector<int> v2;
00649     v.push_back(10);
00650     v.push_back(20);
00651     v2.push_back(10);
00652     v2.push_back(30);
00653
00654     EXPECT_TRUE(v < v2);
00655     EXPECT_TRUE(v2 > v);
00656     EXPECT_FALSE(v > v2);
00657 }
00658
00659 // Test 17: begin() ir end() iteratoriai
00660 TEST_F(VectorTest, BeginEndIterators) {
00661     v.push_back(10);

```

```

00662     v.push_back(20);
00663     v.push_back(30);
00664
00665     int sum = 0;
00666     for (auto it = v.begin(); it != v.end(); ++it) {
00667         sum += *it;
00668     }
00669     EXPECT_EQ(sum, 60);
00670 }
00671
00672 // Test 18: assign()
00673 TEST_F(VectorTest, Assign) {
00674     v.assign(5, 42);
00675     EXPECT_EQ(v.size(), 5);
00676     for (size_t i = 0; i < v.size(); ++i) {
00677         EXPECT_EQ(v[i], 42);
00678     }
00679 }
00680
00681 // Test 19: insert()
00682 TEST_F(VectorTest, Insert) {
00683     v.push_back(10);
00684     v.push_back(30);
00685     v.insert(v.begin() + 1, 20);
00686
00687     EXPECT_EQ(v.size(), 3);
00688     EXPECT_EQ(v[0], 10);
00689     EXPECT_EQ(v[1], 20);
00690     EXPECT_EQ(v[2], 30);
00691 }
00692
00693 // Test 20: erase()
00694 TEST_F(VectorTest, Erase) {
00695     v.push_back(10);
00696     v.push_back(20);
00697     v.push_back(30);
00698
00699     v.erase(v.begin() + 1);
00700
00701     EXPECT_EQ(v.size(), 2);
00702     EXPECT_EQ(v[0], 10);
00703     EXPECT_EQ(v[1], 30);
00704 }
00705
00706 // Test 21: emplace_back()
00707 TEST_F(VectorTest, EmplaceBack) {
00708     v.emplace_back(42);
00709     v.emplace_back(99);
00710
00711     EXPECT_EQ(v.size(), 2);
00712     EXPECT_EQ(v[0], 42);
00713     EXPECT_EQ(v[1], 99);
00714 }
00715
00716 // Test 22: Copy konstruktorius
00717 TEST_F(VectorTest, CopyConstructor) {
00718     v.push_back(10);
00719     v.push_back(20);
00720
00721     Vector<int> v2 = v;
00722     EXPECT_EQ(v2.size(), 2);
00723     EXPECT_EQ(v2[0], 10);
00724
00725     v2[0] = 100;
00726     EXPECT_EQ(v[0], 10); // Original nepakeistas
00727 }
00728
00729 // Test 23: Move konstruktorius
00730 TEST_F(VectorTest, MoveConstructor) {
00731     v.push_back(10);
00732     v.push_back(20);
00733
00734     Vector<int> v2 = std::move(v);
00735     EXPECT_EQ(v2.size(), 2);
00736     EXPECT_EQ(v.size(), 0); // Original tuias
00737 }
00738
00739 // Test 24: Copy assignment
00740 TEST_F(VectorTest, CopyAssignment) {
00741     v.push_back(10);
00742     v.push_back(20);
00743
00744     Vector<int> v2;
00745     v2 = v;
00746     EXPECT_EQ(v2.size(), 2);
00747     v2[0] = 100;
00748     EXPECT_EQ(v[0], 10); // Original nepakeistas

```

```
00749 }
00750
00751 // Test 25: Move assignment
00752 TEST_F(VectorTest, MoveAssignment) {
00753     v.push_back(10);
00754     v.push_back(20);
00755
00756     Vector<int> v2;
00757     v2 = std::move(v);
00758     EXPECT_EQ(v2.size(), 2);
00759     EXPECT_EQ(v.size(), 0); // Original tuias
00760 }
00761
00762 // Test 26: max_size()
00763 TEST_F(VectorTest, MaxSize) {
00764     EXPECT_GT(v.max_size(), 0);
00765 }
00766
00767 // Test 27: data()
00768 TEST_F(VectorTest, Data) {
00769     v.push_back(10);
00770     v.push_back(20);
00771
00772     int* ptr = v.data();
00773     EXPECT_EQ(ptr[0], 10);
00774     EXPECT_EQ(ptr[1], 20);
00775 }
00776
```

skyrius 7

Pavyzdiai

7.1 C:/Users/ditas/source/repos/ditassimoliunas-prog/objektuzdv3/ header_files/Vector.h

Add element to the end of the vector (copy semantics).

Add element to the end of the vector (copy semantics) Appends a copy of the given value to the end. If the current size equals capacity, the vector doubles its capacity and reallocates.

Time Complexity: O(1) amortized

Parametrai

value	The value to append (copied)
-------	------------------------------

limtys

std::bad_alloc	if memory allocation fails
----------------	----------------------------

Taip pat irti

push_back(T&&), pop_back(), emplace_back()

```
Vector<int> v;
v.push_back(42);
v.push_back(100);
#pragma once

#include <iostream>
#include <stdexcept>
#include <utility>
#include <algorithm>
#include <limits>
#include <vector>

/**
 * @brief Random access iterator for Vector<T> container
 *
 * VectorIterator provides bidirectional and random access to Vector elements.
 * Supports all standard iterator operations including comparison, arithmetic,
 * and dereferencing.
 *
 * @tparam T The element type that the iterator points to
 *
 * @see Vector
 */
template <typename T>
class VectorIterator {
public:
    using iterator_category = std::random_access_iterator_tag;
    using value_type = T;
    using difference_type = std::ptrdiff_t;
    using pointer = T*;
```

```

    using reference = T&;

private:
    T* _ptr;

public:
    VectorIterator(T* ptr = nullptr) : _ptr(ptr) {}

    reference operator*() const { return *_ptr; }
    pointer operator->() const { return _ptr; }

    VectorIterator& operator++() {
        ++_ptr;
        return *this;
    }
    VectorIterator operator++(int) {
        VectorIterator temp = *this;
        ++_ptr;
        return temp;
    }

    VectorIterator& operator--() {
        --_ptr;
        return *this;
    }
    VectorIterator operator--(int) {
        VectorIterator temp = *this;
        --_ptr;
        return temp;
    }

    VectorIterator operator+(difference_type n) const {
        return VectorIterator(_ptr + n);
    }
    VectorIterator operator-(difference_type n) const {
        return VectorIterator(_ptr - n);
    }

    VectorIterator& operator+=(difference_type n) {
        _ptr += n;
        return *this;
    }
    VectorIterator& operator-=(difference_type n) {
        _ptr -= n;
        return *this;
    }

    reference operator[](difference_type n) const {
        return _ptr[n];
    }

    bool operator==(const VectorIterator& other) const { return _ptr == other._ptr; }
    bool operator!=(const VectorIterator& other) const { return _ptr != other._ptr; }
    bool operator<(const VectorIterator& other) const { return _ptr < other._ptr; }
    bool operator>(const VectorIterator& other) const { return _ptr > other._ptr; }
    bool operator<=(const VectorIterator& other) const { return _ptr <= other._ptr; }
    bool operator>=(const VectorIterator& other) const { return _ptr >= other._ptr; }

    difference_type operator-(const VectorIterator& other) const {
        return _ptr - other._ptr;
    }
};

/**
 * @class Vector
 * @brief Custom dynamic array container template
 *
 * Vector<T> is a generic container that stores elements of type T in a
 * dynamically allocated array. It provides O(1) amortized time complexity
 * for push_back(), while maintaining contiguous memory layout like std::vector.
 *
 * The class implements the Rule of Five pattern and fully supports deep copy
 * and move semantics.
 *
 * **Key Features:**
 * - Dynamic memory management (automatic resizing)
 * - Random access iterators (begin(), end(), rbegin(), rend())
 * - Copy and move semantics (Rule of Five)
 * - Standard container operations (push_back, pop_back, insert, erase, etc.)
 * - Comparable performance to std::vector for most operations
 *
 * **Memory Strategy:**
 * - When capacity is exhausted, reallocates with roughly 1.5x growth factor
 * - Tracks reallocation count for performance analysis
 * - Supports manual reserve() for optimization
 *
 * **Example Usage:**

```

```

* @code
* Vector<int> v;
* v.push_back(10);
* v.push_back(20);
* v.reserve(100);    // Pre-allocate capacity
*
* for (int elem : v) {
*     std::cout << elem << " ";
* }
* @endcode
*
* @tparam T Element type (must support copy and move semantics)
*
* @see VectorIterator
* @see std::vector
*/
template <typename T>
class Vector {
private:
    T* _data;
    size_t _size;
    size_t _capacity;
    size_t _realloc_count;

    void reallocate(size_t new_capacity) {
        if (new_capacity < _size) {
            throw std::invalid_argument("New capacity must be >= current size");
        }
        T* new_data = new T[new_capacity];
        for (size_t i = 0; i < _size; ++i) {
            new_data[i] = _data[i];
        }
        delete[] _data;
        _data = new_data;
        _capacity = new_capacity;
        ++_realloc_count;
    }

public:
    Vector() : _data(nullptr), _size(0), _capacity(0), _realloc_count(0) {}
    explicit Vector(size_t capacity) : _size(0), _capacity(capacity), _realloc_count(0) {
        _data = capacity > 0 ? new T[capacity] : nullptr;
    }
    Vector(size_t size, const T& value) : _size(size), _capacity(size), _realloc_count(0) {
        _data = size > 0 ? new T[size] : nullptr;
        for (size_t i = 0; i < size; ++i) _data[i] = value;
    }

    // Conversion constructor from std::vector
    Vector(const std::vector<T>& other) : _size(other.size()), _capacity(other.size()), _realloc_count(0) {
        _data = _size > 0 ? new T[_size] : nullptr;
        for (size_t i = 0; i < _size; ++i) _data[i] = other[i];
    }

    ~Vector() { delete[] _data; _data = nullptr; _size = 0; _capacity = 0; }

    Vector(const Vector& other) : _size(other._size), _capacity(other._capacity), _realloc_count(0) {
        _data = _capacity > 0 ? new T[_capacity] : nullptr;
        for (size_t i = 0; i < _size; ++i) _data[i] = other._data[i];
    }
    Vector& operator=(const Vector& other) {
        if (this != &other) {
            delete[] _data;
            _size = other._size;
            _capacity = other._capacity;
            _realloc_count = 0;
            _data = _capacity > 0 ? new T[_capacity] : nullptr;
            for (size_t i = 0; i < _size; ++i) _data[i] = other._data[i];
        }
        return *this;
    }

    Vector(Vector&& other) noexcept : _data(other._data), _size(other._size),
        _capacity(other._capacity), _realloc_count(other._realloc_count) {
        other._data = nullptr;
        other._size = 0;
        other._capacity = 0;
        other._realloc_count = 0;
    }
    Vector& operator=(Vector&& other) noexcept {
        if (this != &other) {
            delete[] _data;
            _data = other._data;
            _size = other._size;
            _capacity = other._capacity;
            _realloc_count = other._realloc_count;
            other._data = nullptr;
        }
    }

```

```

        other._size = 0;
        other._capacity = 0;
        other._realloc_count = 0;
    }
    return *this;
}

inline size_t size() const { return _size; }
inline size_t capacity() const { return _capacity; }
inline bool empty() const { return _size == 0; }

inline T& operator[](size_t index) { return _data[index]; }
inline const T& operator[](size_t index) const { return _data[index]; }

T& at(size_t index) {
    if (index >= _size) {
        throw std::out_of_range("Vector index out of range");
    }
    return _data[index];
}

const T& at(size_t index) const {
    if (index >= _size) {
        throw std::out_of_range("Vector index out of range");
    }
    return _data[index];
}

void clear() {
    _size = 0;
}

T& front() {
    if (_size == 0) {
        throw std::out_of_range("Vector is empty: cannot access front()");
    }
    return _data[0];
}

const T& front() const {
    if (_size == 0) {
        throw std::out_of_range("Vector is empty: cannot access front()");
    }
    return _data[0];
}

T& back() {
    if (_size == 0) {
        throw std::out_of_range("Vector is empty: cannot access back()");
    }
    return _data[_size - 1];
}

const T& back() const {
    if (_size == 0) {
        throw std::out_of_range("Vector is empty: cannot access back()");
    }
    return _data[_size - 1];
}

void reserve(size_t new_capacity) {
    if (new_capacity > _capacity) {
        reallocate(new_capacity);
    }
}

void shrink_to_fit() {
    if (_capacity > _size) {
        if (_size == 0) {
            delete[] _data;
            _data = nullptr;
            _capacity = 0;
        } else {
            reallocate(_size);
        }
    }
}

void resize(size_t new_size) {
    if (new_size > _capacity) {
        reallocate(new_size);
    }
    _size = new_size;
}

void resize(size_t new_size, const T& value) {
    if (new_size > _capacity) {

```



```

        reallocate(new_size);
    }
    if (new_size > _size) {
        for (size_t i = _size; i < new_size; ++i) {
            _data[i] = value;
        }
    }
    _size = new_size;
}

void swap(Vector& other) noexcept {
    std::swap(_data, other._data);
    std::swap(_size, other._size);
    std::swap(_capacity, other._capacity);
    std::swap(_realloc_count, other._realloc_count);
}

bool operator==(const Vector& other) const {
    if (_size != other._size) return false;
    for (size_t i = 0; i < _size; ++i) {
        if (_data[i] != other._data[i]) return false;
    }
    return true;
}

bool operator!=(const Vector& other) const {
    return !(*this == other);
}

bool operator<(const Vector& other) const {
    for (size_t i = 0; i < _size && i < other._size; ++i) {
        if (_data[i] < other._data[i]) return true;
        if (_data[i] > other._data[i]) return false;
    }
    return _size < other._size;
}

bool operator>(const Vector& other) const {
    return other < *this;
}

bool operator<=(const Vector& other) const {
    return !(other < *this);
}

bool operator>=(const Vector& other) const {
    return !(*this < other);
}

using iterator = VectorIterator<T>;
using const_iterator = VectorIterator<const T>;

iterator begin() { return iterator(_data); }
iterator end() { return iterator(_data + _size); }

const_iterator begin() const { return const_iterator(_data); }
const_iterator end() const { return const_iterator(_data + _size); }

const_iterator cbegin() const { return const_iterator(_data); }
const_iterator cend() const { return const_iterator(_data + _size); }

using reverse_iterator = std::reverse_iterator<iterator>;
using const_reverse_iterator = std::reverse_iterator<const_iterator>;

reverse_iterator rbegin() { return reverse_iterator(end()); }
reverse_iterator rend() { return reverse_iterator(begin()); }

const_reverse_iterator rbegin() const { return const_reverse_iterator(end()); }
const_reverse_iterator rend() const { return const_reverse_iterator(begin()); }

const_reverse_iterator crbegin() const { return const_reverse_iterator(end()); }
const_reverse_iterator crend() const { return const_reverse_iterator(begin()); }

void assign(size_t count, const T& value) {
    clear();
    if (count > _capacity) {
        reallocate(count);
    }
    for (size_t i = 0; i < count; ++i) {
        _data[i] = value;
    }
    _size = count;
}

template<typename InputIt>
void assign(InputIt first, InputIt last) {
    // This is a specialized version - only works with Vector iterators

```

```

    // For general case, user should use clear() + for loop
}

iterator insert(const iterator pos, const T& value) {
    size_t index = std::distance(cbegin(), pos);
    if (index > _size) {
        throw std::out_of_range("Iterator out of range");
    }

    if (_size >= _capacity) {
        size_t new_capacity = (_capacity == 0) ? 1 : _capacity * 2;
        reallocate(new_capacity);
    }

    for (size_t i = _size; i > index; --i) {
        _data[i] = _data[i - 1];
    }
    _data[index] = value;
    ++_size;

    return iterator(_data + index);
}

// Non-const overload for insert (accepts non-const iterator, treats as const_iterator)
iterator insert(iterator pos, const T& value) {
    // pos is a mutable iterator, but we just use it as a position
    // Convert to size_t using cbegin()
    size_t index = std::distance(begin(), pos);
    if (index > _size) {
        throw std::out_of_range("Iterator out of range");
    }

    if (_size >= _capacity) {
        size_t new_capacity = (_capacity == 0) ? 1 : _capacity * 2;
        reallocate(new_capacity);
    }

    for (size_t i = _size; i > index; --i) {
        _data[i] = _data[i - 1];
    }
    _data[index] = value;
    ++_size;

    return iterator(_data + index);
}

// Range insert overload with three arguments
template<typename InputIt>
iterator insert(const_iterator pos, InputIt first, InputIt last) {
    size_t index = std::distance(cbegin(), pos);
    if (index > _size) {
        throw std::out_of_range("Iterator out of range");
    }

    // Count elements manually
    size_t count = 0;
    for (InputIt it = first; it != last; ++it) {
        ++count;
    }

    if (_size + count > _capacity) {
        reallocate(std::max(_size + count, _capacity * 2));
    }

    // Shift elements to the right
    for (size_t i = _size; i > index; --i) {
        _data[i + count - 1] = _data[i - 1];
    }

    // Insert new elements
    size_t i = index;
    for (InputIt it = first; it != last; ++it, ++i) {
        _data[i] = *it;
    }
    _size += count;

    return iterator(_data + index);
}

// Range insert overload with non-const iterator
template<typename InputIt>
iterator insert(iterator pos, InputIt first, InputIt last) {
    size_t index = std::distance(begin(), pos);
    if (index > _size) {
        throw std::out_of_range("Iterator out of range");
    }

```

```

    }

    // Count elements manually
    size_t count = 0;
    for (InputIt it = first; it != last; ++it) {
        ++count;
    }

    if (_size + count > _capacity) {
        reallocate(std::max(_size + count, _capacity * 2));
    }

    // Shift elements to the right
    for (size_t i = _size; i > index; --i) {
        _data[i + count - 1] = _data[i - 1];
    }

    // Insert new elements
    size_t i = index;
    for (InputIt it = first; it != last; ++it, ++i) {
        _data[i] = *it;
    }
    _size += count;

    return iterator(_data + index);
}

iterator erase(const_iterator pos) {
    size_t index = std::distance(cbegin(), pos);
    if (index >= _size) {
        throw std::out_of_range("Iterator out of range");
    }

    for (size_t i = index; i < _size - 1; ++i) {
        _data[i] = _data[i + 1];
    }
    --_size;

    return iterator(_data + index);
}

iterator erase(const_iterator first, const_iterator last) {
    size_t first_index = std::distance(cbegin(), first);
    size_t last_index = std::distance(cbegin(), last);

    if (first_index > _size || last_index > _size || first_index > last_index) {
        throw std::out_of_range("Invalid range");
    }

    size_t count = last_index - first_index;
    for (size_t i = first_index; i < _size - count; ++i) {
        _data[i] = _data[i + count];
    }
    _size -= count;

    return iterator(_data + first_index);
}

// Non-const overload for erase (single element)
iterator erase(iterator pos) {
    size_t index = std::distance(begin(), pos);
    if (index >= _size) {
        throw std::out_of_range("Iterator out of range");
    }

    for (size_t i = index; i < _size - 1; ++i) {
        _data[i] = _data[i + 1];
    }
    --_size;

    return iterator(_data + index);
}

// Non-const overload for erase (range)
iterator erase(iterator first, iterator last) {
    size_t first_index = std::distance(begin(), first);
    size_t last_index = std::distance(begin(), last);

    if (first_index > _size || last_index > _size || first_index > last_index) {
        throw std::out_of_range("Invalid range");
    }

    size_t count = last_index - first_index;
    for (size_t i = first_index; i < _size - count; ++i) {
        _data[i] = _data[i + count];
    }
}

```

```

    _size -= count;

    return iterator(_data + first_index);
}

template<typename... Args>
void emplace_back(Args&&... args) {
    if (_size >= _capacity) {
        size_t new_capacity = (_capacity == 0) ? 1 : _capacity * 2;
        reallocate(new_capacity);
    }
    new (_data + _size) T(std::forward<Args>(args)...);
    ++_size;
}

template<typename... Args>
iterator emplace(const_iterator pos, Args&&... args) {
    size_t index = std::distance(cbegin(), pos);
    if (index > _size) {
        throw std::out_of_range("Iterator out of range");
    }

    if (_size >= _capacity) {
        size_t new_capacity = (_capacity == 0) ? 1 : _capacity * 2;
        reallocate(new_capacity);
    }

    for (size_t i = _size; i > index; --i) {
        _data[i] = _data[i - 1];
    }
    new (_data + index) T(std::forward<Args>(args)...);
    ++_size;

    return iterator(_data + index);
}

size_t max_size() const {
    return std::numeric_limits<size_t>::max() / sizeof(T);
}

inline T* data() { return _data; }
inline const T* data() const { return _data; }

inline size_t get_reallocation_count() const { return _realloc_count; }
inline void reset_reallocation_count() { _realloc_count = 0; }

/**
 * @brief Add element to the end of the vector (copy semantics)
 *
 * Appends a copy of the given value to the end. If the current size equals
 * capacity, the vector doubles its capacity and reallocates.
 *
 * **Time Complexity:** O(1) amortized
 *
 * @param value The value to append (copied)
 *
 * @throws std::bad_alloc if memory allocation fails
 *
 * @see push_back(T&&), pop_back(), emplace_back()
 *
 * @example
 * @code
 * Vector<int> v;
 * v.push_back(42);
 * v.push_back(100);
 * @endcode
 */
void push_back(const T& value) {
    if (_size >= _capacity) {
        size_t new_capacity = (_capacity == 0) ? 1 : _capacity * 2;
        reallocate(new_capacity);
    }
    _data[_size] = value;
    ++_size;
}

/**
 * @brief Add element to the end of the vector (move semantics)
 *
 * Appends an rvalue reference to the vector. The element is moved
 * (not copied) into the vector. This is more efficient for expensive-to-copy types.
 *
 * **Time Complexity:** O(1) amortized
 *
 * @param value The rvalue reference to append (moved)
 *
 * @throws std::bad_alloc if memory allocation fails

```

```

*
* @see push_back(const T&), pop_back(), emplace_back()
*
* @example
* @code
* Vector<std::string> v;
* v.push_back(std::string("Hello")); // Moves the temporary
* @endcode
*/
void push_back(T&& value) {
    if (_size >= _capacity) {
        size_t new_capacity = (_capacity == 0) ? 1 : _capacity * 2;
        reallocate(new_capacity);
    }
    _data[_size] = std::move(value);
    ++_size;
}

/**
* @brief Remove the last element from the vector
*
* Decrements the size by one, effectively removing the last element.
* Does nothing if the vector is empty. Note: the element destructor
* is NOT called (similar to std::vector::pop_back for POD types).
*
* **Time Complexity:** O(1) constant
*
* **Precondition:** Empty vector is safe to call on (no-op)
*
* @see push_back(), front(), back()
*
* @example
* @code
* Vector<int> v = {1, 2, 3};
* v.pop_back(); // Now size is 2, last element is 2
* @endcode
*/
void pop_back() {
    if (_size > 0) {
        --_size;
    }
}
};

```


Rodykl

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